Phase space mixing in the presence of self-gravity: spherically symmetric Vlasov-Poisson evolutions from initial data out of equilibrium

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[(Erik) Provisional abstract; to be rewritten when section VI is final.] These notes are devoted to the numerical study of a spherically symmetric, collisionless kinetic gas which is trapped in an external central potential, whose self-gravity is taken into account, and whose initial distribution function is concentrated about one orbital phase, far from any steady state. In the absence of self-gravity, such a gas relaxes in time to a stationary state through phase space mixing. We analyze how this relaxation is modified by the self-gravity of the gas, for the reduced model in which all the gas particles have the same magnitude of the angular momentum, by means of a particle-in-cell code whose particles are placed on a regular lattice in action-angle variables. The outcome is controlled by a parameter μ which compares the amplitude of the potential generated by the gas with the spread of its orbital frequencies. For small μ the gas mixes, more slowly than in the absence of self-gravity, and the potential which it generates relaxes to a stationary profile, as in Landau damping; however, the state which is reached differs from the average of the initial distribution function over the angle variables. For larger μ a compact part of the gas survives as a dense core at the circular orbit, whose radial oscillation is damped. Above a critical value of μ this oscillation persists, and the potential does not relax. For clouds which are compact in angle the critical value lies between 0.43 and 0.48, both in the isochrone and in the Kepler potential.

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I. INTRODUCTION

Collisionless kinetic gases provide the standard description for systems consisting of a large number of particles which interact with each other only through the mean field they generate. Prominent examples are the stars in a galaxy or in a globular cluster, and the dark matter particles in a galactic halo [1]. The state of such a gas at time t is described by its one-particle distribution function (DF) $f(t, x, p)$, a nonnegative function on the one-particle phase space which determines the number of gas particles in a small phase space volume about the point (x, p), and whose evolution is governed by the Vlasov equation.[1] In the Newtonian gravitational case, the Vlasov equation is coupled to Poisson's equation for the gravitational potential generated by the gas, which yields the Vlasov-Poisson system, see Ref. [2] for a review of its mathematical properties. Its general relativistic counterpart is the Vlasov-Einstein system [3].

It is instructive to compare the dynamics of a self-gravitating collisionless gas with the one of a pressureless fluid (dust), which is obtained in the limit in which the velocity dispersion of the gas vanishes. The gravitational collapse of a spherical dust cloud leads to the formation of a singularity in finite time at which the density diverges, both in the Newtonian theory and in general relativity [4]; in the latter case, part of this singularity may even be visible to distant observers [5, 6]. In contrast to this, the velocity dispersion of a kinetic gas prevents the particles from focusing. In fact, the Vlasov-Poisson system admits global classical solutions for general initial data [7–9], while for the spherically symmetric Vlasov-Einstein system global solutions exist for small initial data [10] and sufficiently concentrated data collapse to a black hole [11]. Therefore, for a Newtonian collisionless gas the relevant question is not whether or not a singularity forms, but rather how the gas behaves for large times and, in particular, whether or not it relaxes to a stationary state.

A fundamental mechanism leading to such a relaxation is phase space mixing [12, 13]. To explain it, consider first a gas whose self-gravity is neglected and which is trapped in the potential well of an external potential. If the one-particle Hamiltonian is integrable, each gas particle is confined to an invariant torus, and in terms of action-angle variables (Q, J) the solution of the Vlasov equation is $f(t, Q, J) = F(Q - \Omega(J)t, J)$, where F describes the initial DF and $\Omega(J)$ denotes the frequencies associated with the torus labeled by J. Since these frequencies vary from one torus

[1] The Vlasov equation is also often called the collisionless Boltzmann equation in the literature.

to the other, the DF is stretched along the angle directions and develops structures on finer and finer scales as time increases. As a consequence, f does not converge pointwise as $t \to \infty$. However, the macroscopic observables which are obtained by integrating f against a test function do converge, and their limit is the one associated with the average of the initial DF over the angle variables, which depends only on integrals of motion. Note that this holds for any initial DF, not only for small perturbations of a steady state. A mathematically precise formulation of this statement and its proof for rotationally symmetric potentials in $d = 1, 2, 3$ space dimensions satisfying a suitable non-degeneracy condition were given by Rioseco and Sarbach [14] (see also Refs. [15, 16]). The results in Ref. [14] include the case of a spherically symmetric gas configuration in which all the gas particles have the same value of the total angular momentum, which behaves like an effective one-dimensional system, and they explain the relaxation to a virialized state which had been observed in the numerical simulations of Ref. [17]. The final state is thus determined by the initial DF through its average over the angle variables; for a gas in an axisymmetric potential with two potential wells along the symmetry axis, in which the motion is integrable, this prediction has recently been compared with N-particle simulations in Ref. [18]. For the analogous phenomenon in the exterior of a black hole we refer the reader to Refs. [19, 20], and for estimates on the rate at which the observables converge to Refs. [21–23].

The generalization to the self-gravitating case, which was left open in Ref. [14], is considerably more involved, since the potential now depends on the DF, such that a change of the latter induces a change of the potential which, in turn, acts back on the gas. In plasma physics, where the particles interact through their mean electric field, the corresponding collective effect has been known since the work of Landau [24]: for a gas of charged particles close to a spatially homogeneous equilibrium the electric field decays in time while the DF undergoes mixing. This phenomenon, known as Landau damping, has been established rigorously at the nonlinear level by Mouhot and Villani [25]. The gravitational case differs from the electromagnetic one in two important aspects. First, the interaction is attractive, such that the collective response of the gas tends to reinforce a perturbation instead of screening it. Second, a self-gravitating gas configuration of finite mass does not possess spatially homogeneous equilibria. Instead, its steady states are spatially inhomogeneous, the gas particles are trapped in the potential well, and the frequencies of their radial motion lie in an interval which, in the situations considered in these notes, is bounded and bounded away from zero. For this reason, a collective oscillation whose frequency lies below this interval is not in resonance with any gas particle and hence cannot be damped by mixing.

Two situations have been analyzed in the gravitational case. The first one concerns small perturbations of a self-consistent steady state. Hadžić, Rein and Straub [26] (see also Refs. [27, 28]) showed that the essential spectrum of the operator governing the linearized dynamics about a large class of spherically symmetric steady states is determined by the radial periods of the gas particles, and, using a version of the Birman-Schwinger principle which was first applied to this problem by Mathur [29], they derived a criterion for the existence of an eigenvalue below this spectrum, which corresponds to an oscillating mode that is not damped. Related criteria and estimates for the number of such modes have been given in Ref. [30]. For the reduced system in which all the gas particles have the same magnitude of the angular momentum and move around a central point mass, Hadžić, Rein, Schrecker and Straub [31] proved the following dichotomy for steady states of sufficiently small mass whose DF is proportional to $(E_0 - E)^k$ for energies E below the cutoff energy E_0: the solutions of the linearized system are damped if $k > 1$, whereas an oscillating mode exists if $1/2 < k \leq 1$. Quantitative decay estimates for the linearized system have been obtained in Refs. [23, 32], and particle-in-cell simulations show that, depending on the steady state, its perturbations either oscillate without damping or decay [33, 34]. Perturbations of steady states in the setting of these notes are analyzed in a companion article. [(Erik) Reference to the companion article on perturbations of steady states to be added.] The second situation concerns initial data which are not a perturbation of a steady state of the gas. For a self-gravitating gas around a point mass, Chaturvedi and Luk [35] proved that small, spherically symmetric initial data which are supported on bounded orbits, with angular momenta in an interval bounded away from zero, undergo phase space mixing up to a time of the order of $\epsilon^{-1}(\log \epsilon^{-1})^{-2}$, where ϵ measures the size of the data, while the global existence and the asymptotic behavior of these solutions remain open. Finally, when the initial data are far from any steady state and the self-gravity is strong, the gas undergoes a violent relaxation [13] toward a quasi-stationary state, and the prediction of this state from the initial data is understood only in special cases, see Ref. [36] for a review.

In these notes, we study numerically the relaxation through phase space mixing of a spherically symmetric collisionless gas which is trapped in an external central potential, whose self-gravity is taken into account, and whose initial DF is far from any steady state. We consider the reduced model in which all the gas particles have the same magnitude $L_0 > 0$ of the angular momentum, which has a two-dimensional phase space and is the setting considered in Refs. [14, 31]. This model goes back to the numerical experiments of Hénon with spherical shells (see Ref. [37] and references therein) and has been used in the N-body simulations of Ref. [38]. For the external potential we choose the isochrone potential, for which the action-angle variables are known explicitly, and the potential of a point mass, which is the one considered in Refs. [31, 35]. As initial data we consider a gas cloud whose particles are concentrated about one orbital phase, with a smooth distribution of their action variables. In the absence of self-gravity this cloud mixes, and we address two questions: up to which mass does the gas relax to a stationary state when the self-gravity

of the cloud is taken into account, and which state does it reach. Our analysis is based on a particle-in-cell (PIC) code which solves the nonlinear Vlasov-Poisson system, and on a measure of the mixing in terms of the action-angle variables of the instantaneous potential of the gas.

The main results of these notes can be summarized as follows. First, in the absence of self-gravity the observables relax algebraically, with an exponent which is determined by the behavior of the initial DF at the boundary of its support, and the PIC code reproduces the exact solution of this problem to within 6×10^{-9} in relative terms. Second, when the self-gravity of the gas is taken into account, the observables which measure the mixing have to be computed with the action-angle variables of the total potential; those of the external potential leave a time-independent residue which could be mistaken for an incomplete mixing. Third, the evolution of the cloud is controlled by a parameter μ which compares the amplitude of the potential generated by the cloud with the spread of its orbital frequencies. Below a critical value μ_c the gas relaxes, although more slowly than in the absence of self-gravity: the potential which it generates approaches a stationary profile while the DF continues to mix, which is the analogue of Landau damping. Fourth, the state which is reached is not the average of the initial DF over the angle variables, which is the final state in the absence of self-gravity. Already for a cloud whose mass is one percent of the central mass, the fraction of the gas on nearly circular orbits is more than ten times larger than in that average, and for $\mu \gtrsim 0.4$ a compact part of the cloud survives as a dense core at the circular orbit, whose radial oscillation is damped. Above μ_c the oscillation of the core persists during the time span of our simulations, and the potential does not relax. For clouds which are compact in angle, μ_c lies between 0.43 and 0.48 in the isochrone as well as in the Kepler potential, and the simulations with other widths in action are compatible with this interval, whereas a cloud which is wide in angle requires a larger value. In particular, in the Kepler potential, which is the setting of Ref. [35], a cloud of a given shape only relaxes when its mass lies below a critical value.

Particle methods have a long tradition in the numerical study of spherically symmetric collisionless systems, both in the Newtonian case [33, 34] and in the relativistic one, where they have been used to analyze the collapse of star clusters and the critical behavior at the threshold of black hole formation [39–43]; see Refs. [44–46] for introductions to the PIC method, Ref. [47] for a higher-order implementation in the context of plasma physics, and Ref. [48] for a convergence proof of a PIC scheme for the spherically symmetric Vlasov-Einstein system. Measuring the relaxation of the observables with such a method is delicate, because the signal has to be separated from the noise which is due to the finite number of computational particles. In these notes we address this problem by placing the particles on a regular lattice in action-angle variables, such that the sums over the particles are quadrature rules for the corresponding integrals over phase space. A loading of the particles which reduces the initial fluctuations (a quiet start) was already used for systems of spherical shells in Ref. [37]. Since the gas does not relax with respect to a fixed potential, we compute the observables with the action-angle variables of the instantaneous potential.

The remainder of these notes is organized as follows. In section II we formulate the Vlasov-Poisson system with an external potential in spherical symmetry and the reduced model with fixed angular momentum, and we introduce the action-angle variables and the observables used throughout these notes. In section III we review the mixing in the absence of self-gravity. Our numerical methods are described in section IV, and their validation is presented in section V. Section VI is devoted to the evolution of the gas cloud. Conclusions and an outlook, including the extension to a distribution of angular momenta, are given in section VII. Technical details regarding the numerical computation of the action-angle variables are included in Appendix A.

II. THE VLASOV-POISSON SYSTEM IN SPHERICAL SYMMETRY

In this section we formulate the Vlasov-Poisson system with an external potential and reduce it to spherical symmetry, first for a gas whose particles have arbitrary angular momenta and then for the reduced model in which all of them have the same magnitude of the angular momentum. Next, we introduce the dimensionless variables used in our numerical computations, the action-angle variables associated with a time-independent potential, in terms of which we characterize the steady states, and, finally, the observables on which our analysis of the mixing phenomenon is based.

We consider a collisionless gas consisting of identical particles which are subject to a time-independent external potential Φ_{ext} and to the gravitational potential Φ_{gas} generated by the gas itself. Since all the gas particles have the same mass, it is convenient to normalize the DF such that $f(t, x, p) d^3x\, d^3p$ is the mass of the gas contained in the volume element $d^3x\, d^3p$ of phase space, where from now on p denotes the momentum per unit mass.[2] With these

[2] This is equivalent to working in units for which the mass of each gas particle is one, as in Ref. [14].

conventions, the Vlasov-Poisson system consists of the Vlasov equation

$$\frac{\partial f}{\partial t} + p \cdot \nabla_x f - (\nabla_x \Phi) \cdot \nabla_p f = 0, \qquad \Phi := \Phi_{ext} + \Phi_{gas}, \tag{1}$$

and the Poisson equation

$$\Delta \Phi_{gas} = 4\pi G \rho, \qquad \rho(t,x) := \int\limits_{\mathbb{R}^3} f(t,x,p) d^3p, \tag{2}$$

where G denotes Newton's constant and ρ is the mass density of the gas. To obtain a unique solution of Eq. (2) we require that $\Phi_{gas}(t,x) \to 0$ for $|x| \to \infty$.

For the external potential we choose Hénon's isochrone potential [1, 49, 50]

$$\Phi_{ext}(x) = -\frac{GM}{b + \sqrt{b^2 + r^2}}, \qquad r := |x|, \tag{3}$$

which is generated by a spherical matter distribution of total mass M. The parameter $b \geq 0$ is a length scale which describes the transition from a harmonic potential (when $r \ll b$) to the potential of a point particle of mass M (when $r \gg b$). For $b = 0$, Eq. (3) reduces to the Kepler potential $\Phi_{ext} = -GM/r$, which is the external potential considered in Refs. [31, 35]. The reason for this choice is that in this potential the period of the radial motion depends only on the energy of the particle [49, 50], and that the corresponding action-angle variables are known in closed form [1]. When the self-gravity of the gas is neglected, this provides exact solutions against which our numerical methods can be validated.

A. Reduction to spherical symmetry

A spherically symmetric gas configuration is described by a DF which is invariant with respect to simultaneous rotations of x and p. Such a DF depends on (x,p) only through the radius r, the radial component $p_r := x \cdot p/r$ of the momentum and the magnitude $L := |x \wedge p|$ of the angular momentum, that is,

$$f(t,x,p) = \mathcal{F}(t,r,p_r,L). \tag{4}$$

In this case, the density ρ and the potential Φ_{gas} depend only on t and r, and the Vlasov equation (1) reduces to

$$\frac{\partial \mathcal{F}}{\partial t} + p_r \frac{\partial \mathcal{F}}{\partial r} - \frac{\partial V_L}{\partial r} \frac{\partial \mathcal{F}}{\partial p_r} = 0, \qquad V_L(t,r) := \Phi(t,r) + \frac{L^2}{2r^2}, \tag{5}$$

with V_L the effective potential for the radial motion. Since no derivative with respect to L appears in Eq. (5), the gas particles with a given value of L behave like a one-dimensional gas in the potential V_L. However, particles with different values of L interact with each other through the potential Φ_{gas}, which is determined by the spherically reduced Poisson equation

$$\frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial \Phi_{gas}}{\partial r} \right) = 4\pi G \rho, \qquad \rho(t,r) = \frac{2\pi}{r^2} \int\limits_0^\infty \int\limits_{-\infty}^\infty \mathcal{F}(t,r,p_r,L)\, dp_r\, L dL, \tag{6}$$

see for instance Ref. [17]. The solution of Eq. (6) which is regular at the center and vanishes at infinity is

$$\Phi_{gas}(t,r) = -4\pi G \int\limits_0^\infty \frac{\rho(t,s)}{\max\{r,s\}} s^2 ds, \tag{7}$$

and its gradient is $\partial \Phi_{gas}/\partial r = G m(t,r)/r^2$, where $m(t,r) := 4\pi \int_0^r \rho(t,s) s^2 ds$ is the mass of the gas contained in the sphere of radius r at time t.

For the following, it is useful to note that when a spherically symmetric function is integrated over phase space, the volume element can be replaced with

$$d^3x\, d^3p = 8\pi^2 dr\, dp_r\, L dL. \tag{8}$$

In particular, the total mass and the total energy of the gas are given by

$$M_{gas} = 8\pi^2 \int_0^\infty \int_{-\infty}^\infty \int_0^\infty \mathcal{F}(t,r,p_r,L)\, dr\, dp_r\, LdL, \tag{9}$$

$$\mathcal{E}_{gas} = 8\pi^2 \int_0^\infty \int_{-\infty}^\infty \int_0^\infty \mathcal{F}(t,r,p_r,L) \left[\frac{p_r^2}{2} + \frac{L^2}{2r^2} + \Phi_{ext}(r) + \frac{1}{2}\Phi_{gas}(t,r)\right] dr\, dp_r\, LdL, \tag{10}$$

both of which are conserved in time. The conservation of $\mathcal{E}_{gas}$ provides a test for the accuracy of our numerical evolutions, see section V.

B. The reduced model with fixed angular momentum

A further simplification, which has been considered in Refs. [14, 17, 31], is obtained by assuming that all the gas particles have the same magnitude $L_0 > 0$ of the angular momentum.[3] In this case, the DF is of the form

$$\mathcal{F}(t,r,p_r,L) = \mathcal{F}_0(t,r,p_r)\delta(L - L_0), \tag{11}$$

and Eqs. (5,6) yield

$$\frac{\partial \mathcal{F}_0}{\partial t} + p_r \frac{\partial \mathcal{F}_0}{\partial r} - \frac{\partial V_{L_0}}{\partial r}\frac{\partial \mathcal{F}_0}{\partial p_r} = 0, \qquad \rho(t,r) = \frac{2\pi L_0}{r^2} \int_{-\infty}^\infty \mathcal{F}_0(t,r,p_r)\, dp_r, \tag{12}$$

where the potential Φ_{gas} is still given by Eq. (7). Likewise, Eqs. (9,10) hold with $\mathcal{F}$ replaced by $\mathcal{F}_0$, L by L_0, and the integral over L by the factor L_0. Equations (12,7) constitute a Vlasov-Poisson system on a two-dimensional phase space with coordinates (r, p_r). It describes a gas of concentric spherical shells which move in the effective potential V_{L_0}. The centrifugal term in this potential prevents the shells from reaching the center, and the gas configurations considered in these notes are confined to a region of the form $0 < r_{in} \le r \le r_{out}$. In spite of its simplicity, the reduced model retains the two ingredients which are relevant for these notes, namely the dependency of the frequency of the radial motion on the orbit, which is responsible for the mixing, and the self-gravity of the gas.

C. Dimensionless variables

In our numerical computations we use dimensionless variables, which are obtained by measuring masses in units of M, lengths in units of b and times in units of the dynamical time

$$t_{dyn} := \sqrt{\frac{b^3}{GM}}. \tag{13}$$

Accordingly, p_r is measured in units of $\sqrt{GM/b}$, L in units of $\sqrt{GMb}$, the potentials and the energy per unit mass in units of GM/b, and $\mathcal{F}$ in units of $M/(GMb)^{3/2}$. In terms of these variables, all the equations of this section retain their form with $G = M = b = 1$; in particular, the isochrone potential is $\Phi_{ext}(r) = -1/(1 + \sqrt{1 + r^2})$. The resulting system contains no free parameters, and the strength of the self-gravity is controlled by the ratio M_{gas}/M between the mass of the gas and the mass which generates the external potential. In the limit $M_{gas}/M \to 0$ the potential Φ_{gas} can be neglected, and one recovers the model analyzed in Ref. [14].

In the Kepler case $b = 0$ the external potential does not provide a length scale. However, in this case the system (5,6) is invariant with respect to the rescaling

$$(t,r,p_r,L,\mathcal{F},\Phi) \mapsto \left(\alpha^{3/2}t, \alpha r, \alpha^{-1/2}p_r, \alpha^{1/2}L, \alpha^{-3/2}\mathcal{F}, \alpha^{-1}\Phi\right), \qquad \alpha > 0, \tag{14}$$

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[3] Note that in Ref. [31] the symbol L refers to the square of the magnitude of the angular momentum.

which leaves M_{gas} unchanged. Therefore, b can be replaced with an arbitrary length in the definitions above, such that $\Phi_{ext}(r) = -1/r$, and in the reduced model this freedom can be used to fix the value of L_0.

To give an idea of the physical time scale, we note that $M = 10^5 M_\odot$ and $b = 1$ pc, which are of the order of magnitude of the mass and of the core radius of a globular cluster [1], yield $t_{dyn} \approx 4.7 \times 10^4$ yr. As will be seen below, $2\pi t_{dyn}$ is a lower bound for the periods of the radial motion in the isochrone potential.

D. Action-angle variables and steady states

Suppose that the total potential Φ is independent of time. In our applications, Φ is the external potential (3), the total potential Φ_{eq} of a self-consistent steady state (defined below), or the potential of an evolving gas at a fixed instant of time, see section II E. In all these cases $\Phi'(r) = Gm_{tot}(r)/r^2$, where $m_{tot}(r)$ denotes the total mass contained in the sphere of radius r, including the one which generates the external potential. The energy per unit mass

$$E := \frac{p_r^2}{2} + V_L(r) \tag{15}$$

is conserved along the trajectories of the gas particles, and following Refs. [1, 14] we introduce action-angle variables for their radial motion. Since the function $r \mapsto rm_{tot}(r)$ increases strictly monotonically from 0 to ∞, for each $L > 0$ the effective potential V_L has a unique critical point $r = r_{min}(L)$. It is determined by the equation $r^3\Phi'(r) = L^2$, it is a non-degenerate minimum of V_L, and it corresponds to a circular orbit with energy $E_{min}(L) := V_L(r_{min}(L))$. Since $V_L(r) \to \infty$ for $r \to 0$ and $V_L(r) \to 0$ for $r \to \infty$, the orbits with energies in the range $E_{min}(L) < E < 0$ are bound, and they oscillate between the turning points $r_1(E, L) < r_2(E, L)$, which are the two solutions of the equation $V_L(r) = E$.

As in Ref. [14] we define the area function

$$A(E, L) := \oint p_r dr = 2 \int_{r_1(E,L)}^{r_2(E,L)} \sqrt{2(E - V_L(r))}\, dr, \tag{16}$$

which represents the area enclosed by the orbit in the (r, p_r)-plane, and whose partial derivative with respect to E yields the period of the radial motion,

$$T(E, L) := \frac{\partial A}{\partial E}(E, L) = 2 \int_{r_1(E,L)}^{r_2(E,L)} \frac{dr}{\sqrt{2(E - V_L(r))}}. \tag{17}$$

The action and angle variables associated with the radial motion are

$$J := \frac{A(E, L)}{2\pi}, \qquad Q := \frac{2\pi}{T(E, L)} \int_{\gamma_r} \frac{dr}{p_r}, \tag{18}$$

where γ_r denotes the segment of the orbit in the (r, p_r)-plane which connects, in the clockwise direction, the inner turning point $(r_1(E, L), 0)$ with the given point (r, p_r). Hence, Q/Ω with $\Omega := 2\pi/T(E, L)$ is the time which has elapsed since the last passage of the particle through its pericenter; in particular, $Q = 0$ at the pericenter, $Q = \pi$ at the apocenter, and $0 < Q < \pi$ when $p_r > 0$. For each fixed value of L, the map $(r, p_r) \mapsto (Q, J)$ is symplectic, that is $dr \wedge dp_r = dQ \wedge dJ$, and it provides smooth coordinates on the set of bound orbits which are not circular. Since $T > 0$, the first relation in Eq. (18) can be inverted for each L, which yields the energy as a function $E(J, L)$ of the action variable and the angular momentum, and the frequency of the radial motion is

$$\Omega(J, L) = \frac{\partial E}{\partial J}(J, L) = \frac{2\pi}{T(E, L)}. \tag{19}$$

[(Erik) We denote the orbital frequency by Ω (it is called ω in Ref. [14]) and reserve ω for the frequency of the collective oscillation.] In terms of the variables (Q, J, L), the volume element (8) is $8\pi^2 dQ\, dJ\, LdL$, and the Vlasov equation (5) for a time-independent potential assumes the simple form

$$\frac{\partial \mathcal{F}}{\partial t} + \Omega(J, L)\frac{\partial \mathcal{F}}{\partial Q} = 0. \tag{20}$$

In particular, the time-independent solutions of Eq. (20) are the DFs which do not depend on the angle variable. Since $E(J,L)$ is a strictly increasing function of J, they are of the form

$$\mathcal{F}(r,p_r,L)=\psi(E,L), \tag{21}$$

with a nonnegative function ψ. Such a DF is a solution of the Vlasov-Poisson system if the potential Φ_{gas}, which enters the definition of E through V_L, is the one generated by the density of the DF through Eqs. (6,7). This is a nonlinear problem for Φ_{gas}. A solution $(\mathcal{F}_{eq},\Phi_{eq})$ of this problem, with Φ_{eq} the total potential and ρ_{eq} the corresponding density, will be referred to as a self-consistent steady state.

For the external potential (3) the area function can be computed explicitly, see Refs. [1, 14]. In terms of the dimensionless variables introduced in the previous subsection, one finds

$$A(E,L)=2\pi\left[\frac{1}{\sqrt{-2E}}-c(L)\right],\qquad E_{min}(L)=-\frac{1}{2c(L)^2}, \tag{22}$$

with $c(L):=\left(L+\sqrt{L^2+4}\right)/2$ for the isochrone potential and $c(L):=L$ for the Kepler potential. This yields

$$E(J,L)=-\frac{1}{2\left[J+c(L)\right]^2},\qquad \Omega(J,L)=\frac{1}{\left[J+c(L)\right]^3}=(-2E)^{3/2}. \tag{23}$$

Therefore, the frequency depends only on the energy, it decreases monotonically when J or L increase, and for each fixed L its maximum is attained in the limit $J\to 0$ of circular orbits. For the isochrone potential, $c(L)>1$ implies that $\Omega<1$, such that the periods of the radial motion are larger than 2π. An explicit expression for the angle variable Q in terms of (r,p_r,L) can be found in section 3.5.2 of Ref. [1]; its inversion requires the solution of an equation of Kepler's type. In contrast to this, when the self-gravity of the gas is taken into account the potential is only known numerically, and in this case we compute the action-angle variables by numerical quadrature of Eqs. (16,17,18), see Appendix A.

E. Observables

As in Ref. [14], we describe the macroscopic state of the gas through observables of the form

$$N_\varphi(t):=\int f(t,x,p)\varphi(x,p)\,d^3x\,d^3p, \tag{24}$$

where φ is a given test function on phase space. In these notes we consider two classes of such observables.

The first one refers to the action-angle variables (Q,J) belonging to a time-independent potential. When the self-gravity of the gas is neglected, this is the external potential. When it is taken into account and the gas is far from equilibrium, there is no fixed potential with respect to which the gas relaxes, and we use the action-angle variables of the instantaneous potential: at each time t, (Q,J) are the action-angle variables which belong to the potential $r\mapsto\Phi(t,r)$, regarded as a time-independent potential. For a spherically symmetric configuration and a test function of the form $\varphi=a(Q)B(J,L)$, with real-valued functions a and B, Eqs. (8,24) and the expansion of a in a Fourier series yield

$$N_\varphi(t)=\sum_{n\in\mathbb{Z}}\hat{a}_n^*\,h_n(t),\qquad \hat{a}_n:=\frac{1}{2\pi}\int_0^{2\pi}a(Q)e^{-inQ}dQ, \tag{25}$$

with the functions

$$h_n(t):=8\pi^2\int_0^\infty\int_0^\infty\int_0^{2\pi}\mathcal{F}(t,Q,J,L)B(J,L)e^{-inQ}\,dQ\,dJ\,LdL,\qquad n\in\mathbb{Z}. \tag{26}$$

In the reduced model, B depends only on J and Eq. (26) holds with $\mathcal{F}$ replaced by $\mathcal{F}_0$ and the integral over L by the factor L_0. Since $\mathcal{F}$ is real, $h_{-n}=h_n^*$, such that it is sufficient to consider $n\geq 0$. For a steady state h_n vanishes for all $n\neq 0$, whereas h_0 is equal to the mass of the gas weighted with the function B. Therefore, the functions $h_n(t)$ with $n\neq 0$ measure the departure of the gas configuration from a steady state, and their decay in time is the signature of the relaxation process we are interested in. We stress that this property relies on the use of the action-angle variables

which belong to the total potential. If those of the external potential are used instead, a gas whose self-gravity is not negligible gives rise to nonvanishing values of h_n with $n \neq 0$ even after it has relaxed, as we show in section V. When the action-angle variables of the instantaneous potential are used, the test function depends on the DF through $\Phi(t, \cdot)$, such that h_n is no longer a linear functional of the DF. On the other hand, h_n with $n \neq 0$ then vanishes for any steady state, including a steady state which the gas may approach at late times and which is not known in advance.

The second class of observables consists of the deviation of the gravitational potential from its average $\langle\Phi\rangle(r)$ over a late time interval, which is specified in section VI,

$$\delta\Phi(t,r) := \Phi(t,r) - \langle\Phi\rangle(r) = -4\pi G \int_0^\infty \frac{\rho(t,s) - \langle\rho\rangle(s)}{\max\{r,s\}} s^2 ds, \tag{27}$$

where we have used Eq. (7) and $\langle\rho\rangle$ denotes the average of the density over the same interval. For each fixed r, $\Phi_{gas}(t,r)$ is the observable (24) with the test function $\varphi(x,p) = -G/\max\{r,|x|\}$. In contrast to the functions $h_n(t)$, the quantity $\delta\Phi$ does not refer to any particular coordinates on phase space. The results in Refs. [31, 32] refer to the decay of the force field $-\partial\delta\Phi/\partial r$ which is generated by it, or of the time derivative of this field. To measure its magnitude we use the root mean square

$$\|\delta\Phi(t)\| := \left(\frac{1}{r_b - r_a} \int_{r_a}^{r_b} \delta\Phi(t,r)^2 dr \right)^{1/2} \tag{28}$$

over an interval $[r_a, r_b]$ which covers the region occupied by the gas.

III. PHASE SPACE MIXING IN THE ABSENCE OF SELF-GRAVITY

In this section we review the mixing phenomenon in the absence of self-gravity, for an arbitrary initial DF, and discuss the rate at which the observables introduced in the previous section decay. The derivations are formal; for rigorous statements we refer the reader to Refs. [14, 16].

When the self-gravity of the gas is neglected, $\Phi = \Phi_{ext}$ is independent of time, and the solution of Eq. (20) is

$$\mathcal{F}(t,Q,J,L) = F(Q - \Omega(J,L)t, J, L), \tag{29}$$

where the function F, which is 2π-periodic in its first argument, describes the initial DF. No assumption on F is needed here: it need not be close to a steady state. Expanding F in a Fourier series with respect to the angle variable,

$$F(Q,J,L) = \sum_{n\in\mathbb{Z}} \hat{F}_n(J,L) e^{inQ}, \qquad \hat{F}_n(J,L) := \frac{1}{2\pi} \int_0^{2\pi} F(Q,J,L) e^{-inQ} dQ, \tag{30}$$

and introducing Eq. (29) into Eq. (26) one obtains

$$h_n(t) = 16\pi^3 \int_0^\infty \int_0^\infty \hat{F}_n(J,L) B(J,L) e^{-in\Omega(J,L)t}\, dJ\, L dL. \tag{31}$$

Hence, h_0 is constant in time, whereas for $n \neq 0$ the function $h_n(t)$ is a superposition of oscillations whose frequencies $n\Omega(J,L)$ vary from one orbit to the other. Provided that the gradient of Ω with respect to (J,L) is different from zero for almost all (J,L) (in the reduced model, provided that $\partial\Omega/\partial J \neq 0$ for almost all J), it follows from the Riemann-Lebesgue lemma and its generalizations (cf. Refs. [14, 16]) that $h_n(t) \to 0$ for $t \to \infty$ and all $n \neq 0$. As a consequence, the observables (25) converge to the ones which belong to the angle-averaged DF $\hat{F}_0(J,L)$, which is a steady state according to the remark below Eq. (20). This is the phase space mixing property, and $\hat{F}_0$ is the final state of the gas. For the isochrone and the Kepler potentials, Eq. (23) gives $\partial\Omega/\partial J = -3[J + c(L)]^{-4} < 0$, such that the non-degeneracy condition is satisfied.

The rate at which $h_n(t)$ decays is not universal. It depends on the regularity of the function $\hat{F}_n B$ and on its behavior at the boundary of its support. If this function is smooth and vanishes together with all its derivatives at the boundary, repeated integration by parts in Eq. (31) shows that $h_n(t)$ decays faster than any inverse power of t. A

different situation arises in the reduced model when the DF has its support in an interval $0 \leq J \leq J_{max}$ and vanishes like a power at the outer edge, such that

$$\hat{F}_n(J)B(J) = (J_{max} - J)^k g(J) \tag{32}$$

with an exponent $k > 0$ and a smooth function g satisfying $g(J_{max}) \neq 0$. Suppose that $\Omega'(J) < 0$ for all $0 < J \leq J_{max}$, as is the case in all the examples of these notes, and that g vanishes sufficiently fast for $J \to 0$. Then, the behavior of the integral in Eq. (31) for large times is determined by the endpoint $J = J_{max}$ (see, for instance, chapter 6 in Ref. [51]), and for $n \geq 1$ one finds

$$h_n(t) \simeq 16\pi^3 L_0 \, \Gamma(k+1) \, g(J_{max}) \frac{e^{-in\Omega_{min}t}}{[in|\Omega'(J_{max})|t]^{k+1}}, \qquad t \to \infty, \tag{33}$$

where $\Omega_{min} := \Omega(J_{max})$ is the frequency of the outermost orbit and Γ denotes the Gamma function. Therefore, an edge at which the DF vanishes with the exponent k gives rise to an algebraic tail which is proportional to $t^{-(k+1)}$ and which oscillates with the frequency $n\Omega_{min}$: the less regular the DF is at the edge, the slower is the decay. The general rule behind this result is that a singularity of the form $(\omega - \omega_0)^a$ of the Laplace transform of an observable (with a logarithmic factor when a is an integer) gives rise to a tail which is proportional to $t^{-(a+1)}$ and oscillates with the frequency ω_0; the singularities which occur for systems with one and with two action variables have been classified in Refs. [52, 53]. Decay estimates for the mixing in the potential of a steady state whose DF vanishes like a power at the edge of its support have been proven in Ref. [23].

When the angular momenta of the gas particles are distributed, the integration over L in Eq. (31) provides an additional source of decay, since Ω depends on L as well. For the isochrone and the Kepler potentials, Eq. (23) yields $\partial\Omega/\partial L = c'(L)\,\partial\Omega/\partial J$ with $1/2 < c'(L) \leq 1$, such that a dispersion in L is almost as efficient for the mixing as a dispersion of the same magnitude in J. In these notes we restrict ourselves to the reduced model.

When the self-gravity of the gas is taken into account, the potential depends on time, and there is no longer a fixed set of action-angle variables in which the motion of the gas particles is trivial. If the initial DF is close to a self-consistent steady state, the deviation from that state can be described by the equations linearized about it, as in the works on Landau damping and oscillating modes cited in section I. The initial data of these notes are far from any steady state, and their evolution has to be followed with the nonlinear Vlasov-Poisson system, which we solve with the PIC method described in the next section. In this case we compute the observables (26) with the action-angle variables of the instantaneous potential, see section II E.

IV. NUMERICAL METHODS

In this section we describe the PIC method which we use to solve the nonlinear Vlasov-Poisson system, and the representation of the initial data by the computational particles.

A. Particle-in-cell method

In the PIC method [44, 45] the DF is represented by a finite number N of computational particles, each of which stands for a large number of gas particles. In spherical symmetry a computational particle is a spherical shell, which is characterized by its radius $r_j(t)$, its radial momentum $p_j(t)$, its angular momentum L_j and its mass m_j, $j = 1, 2, \ldots, N$, the last two quantities being constant in time. This corresponds to the approximation

$$\mathcal{F}(t, r, p_r, L) \approx \sum_{j=1}^{N} \frac{m_j}{8\pi^2 L_j} \, \delta(r - r_j(t)) \, \delta(p_r - p_j(t)) \, \delta(L - L_j) \tag{34}$$

of the DF, where the factor $8\pi^2 L_j$ is due to the volume element (8). Introducing Eq. (34) into the Vlasov equation (5) yields the equations of motion

$$\frac{dr_j}{dt} = p_j, \qquad \frac{dp_j}{dt} = \frac{L_j^2}{r_j^3} - \frac{\partial\Phi_{ext}}{\partial r}(r_j) - \frac{\partial\Phi_{gas}}{\partial r}(t, r_j) \tag{35}$$

for the computational particles; in the reduced model $L_j = L_0$ for all j. The first two terms on the right-hand side of the second equation are evaluated analytically, whereas the last one requires the solution of the Poisson equation (6). To this purpose we introduce the uniform grid $R_i := (i - 1/2)\Delta r$, $i = 1, 2, \ldots, N_r$, which is staggered with respect to the center, and proceed in four steps.

(i) The mass of each computational particle is assigned to its two neighboring grid points with the linear weights of the cloud-in-cell scheme [44, 45], such that the grid point R_i receives the mass

$$\mu_i := \sum_{j=1}^{N} m_j W\left(\frac{R_i - r_j}{\Delta r}\right), \qquad W(y) := \max\{1 - |y|, 0\}. \tag{36}$$

(ii) The density is approximated by the continuous, piecewise linear function $\rho(t, r)$ which assumes the values

$$\rho_i := \frac{\mu_i}{4\pi\Delta r\,(R_i^2 + \Delta r^2/6)} \tag{37}$$

at the grid points and which is constant for $0 \le r \le R_1$. The denominator in Eq. (37) is the integral of the function $4\pi r^2 W((r - R_i)/\Delta r)$ over r, which guarantees that the mass of the reconstructed density coincides with the total mass of the computational particles.

(iii) The mass function $m(t, r)$ and the potential are obtained by integrating the equations $\partial m/\partial r = 4\pi r^2\rho$ and $\partial\Phi_{gas}/\partial r = Gm/r^2$ outwards from the center. Since ρ is linear on each interval $[R_{i-1}, R_i]$, both integrals are computed in closed form, such that no further discretization error is introduced. The additive constant in Φ_{gas} is fixed by requiring that $\Phi_{gas} + r\,\partial\Phi_{gas}/\partial r = 0$ at the outermost grid point, which matches the potential to the exterior solution $-GM_{gas}/r$.

(iv) The potential and its gradient are interpolated from the grid points to the positions of the computational particles with the same weights as in Eq. (36).

For particles which lie within the distance Δr from the center, the weights in steps (i) and (iv) are complemented by those of their mirror images at $-r_j$, in accordance with the regularity of the density at the center, and beyond the outermost grid point the exterior solution is used. Neither of these situations plays a role for the configurations considered in these notes, in which the gas occupies a shell that lies well inside the grid; the computational particles which leave the grid for the widest initial data of section VI carry less than 10^{-11} of the mass of the gas. For the Kepler potential the code uses the potential of a homogeneous sphere of mass M and unit radius, which coincides with the Kepler potential for $r \ge 1$. In all the simulations of these notes with this potential the gas remains in the region $r > 1.7$, such that the two potentials are equivalent.

Note that the potential which acts on a computational particle includes the contribution of the particle itself. In PIC schemes based on Cartesian coordinates the use of the same weights for the assignment and for the interpolation guarantees that a particle exerts no force on itself [44, 45]. In spherical symmetry this is not the case, and it should not be: a thin shell of mass m and radius r is subject to the mean value of the gravitational fields on its two sides, that is, to the force $-Gm/(2r^2)$ per unit mass. This is the self-gravity of the shell; the scheme reproduces it up to a correction of the order $\Delta r/r$, and we do not subtract it. In fact, we have verified that subtracting it reduces the order of convergence with respect to $1/N$ from two to one in the collapse of a cold homogeneous sphere, for which the exact solution is known.

Equations (35) are integrated in time with the fourth-order symplectic method of Yoshida [54], which is a composition of three steps of the second-order leapfrog method with step sizes $w_1\Delta t$, $w_0\Delta t$ and $w_1\Delta t$, where $w_1 := 1/(2 - 2^{1/3})$ and $w_0 := 1 - 2w_1$. A leapfrog step with step size h consists of the update $p_j \mapsto p_j + hf_j/2$ of the momenta, with f_j the right-hand side of the second equation in (35), followed by the update $r_j \mapsto r_j + hp_j$ of the radii, the computation of the potential for the new radii, and a second update of the momenta with the new values of f_j. Hence, the Poisson equation is solved three times per time step. We use a symplectic method because, for the motion in a given potential, it does not introduce a secular drift of the energy and of the action variables of the particles, which would otherwise limit the accuracy of the observables at late times. For the same reason we keep the time step Δt fixed.

At selected times we compute the total energy (10),

$$\mathcal{E}_{gas} \approx \sum_{j=1}^{N} m_j \left[\frac{p_j^2}{2} + \frac{L_j^2}{2r_j^2} + \Phi_{ext}(r_j) + \frac{1}{2}\Phi_{gas}(t, r_j)\right], \tag{38}$$

the potential Φ_{gas} on the grid, from which $\delta\Phi$ is obtained according to Eq. (27), and the functions

$$h_n(t) \approx \sum_{j=1}^{N} m_j B(J_j, L_j) e^{-inQ_j}, \tag{39}$$

which follow from Eqs. (26,34). Here (Q_j, J_j) denote the action-angle variables of the particle j at the time t with respect to the potential specified in section II E. They are computed from (r_j, p_j) with the method described in Appendix A; for the instantaneous potential, Φ_{gas} is the potential computed by the PIC code on its grid at the time t. In the tests of section V we choose the weight function

$$B(J) = J^2 \exp\left[-\frac{(J-J_1)^2}{\sigma_1^2}\right], \tag{40}$$

with parameters J_1 and σ_1 which are specified below. The factor J^2 makes the function $B(J)e^{-inQ}$ continuous at the circular orbit for $|n| \leq 4$. For the evolution of the gas cloud in section VI we choose $B = 1$, such that h_n/h_0 is the average of e^{-inQ} over the mass of the gas.

The simulations of these notes use an implementation of this method for the reduced model. Unless otherwise stated, they use $\Delta r = 0.1$, a grid which extends to $r = 20$, a time step $\Delta t = 0.05$ or 0.1, and $N = 10^4$ computational particles. [(Erik) To be decided: whether the codes and the scripts which reproduce the results are made public, in which case the repository should be cited here.]

B. Initial data

The initial data considered in these notes are given as functions of the action-angle variables (Q, J) of the external potential $\Phi_0 := \Phi_{ext}$.

If the radii and momenta of the computational particles were drawn at random from the initial DF, the observables would be affected by fluctuations of the order of $N^{-1/2}$. For the numbers of particles used in this work, these fluctuations would exceed the values of $|h_n|/h_0$ which are reached during the mixing of the gas. For this reason we place the computational particles on a regular lattice in the variables (Q, J),

$$J_a := \left(a - \frac{1}{2}\right)\frac{J_{max}}{N_J}, \quad a = 1, \ldots, N_J, \qquad Q_b := \left(b - \frac{1}{2}\right)\frac{2\pi}{N_Q}, \quad b = 1, \ldots, N_Q, \tag{41}$$

where $[0, J_{max}]$ is the range of the action variable which is covered by the DF, such that $N = N_J N_Q$. The radii and momenta of the particles are obtained by inverting the map $(r, p_r) \mapsto (Q, J)$ which belongs to Φ_0, which only requires the solution of an equation of Kepler's type (see below), and the particle at the lattice point (Q_b, J_a) is given a mass m_{ab} which is proportional to the value of the initial DF at this point, with a constant of proportionality which is determined by the total mass M_{gas}. Since the map $(r, p_r) \mapsto (Q, J)$ is symplectic, all the cells of the lattice have the same volume in phase space. Hence, the sums over the computational particles in Eqs. (38,39) are midpoint rules for the corresponding integrals over phase space, and they are free of the fluctuations of a random sample. In plasma physics, a loading of the particles which avoids these fluctuations is known as a quiet start [44]. The method is only reliable over a finite time span: in the absence of self-gravity the rows of the lattice produce a spurious recurrence of the functions h_n, which is quantified in section V, and in its presence the regularity of the lattice is eventually lost due to the discreteness of the particles.

As initial datum we consider the gas cloud

$$\mathcal{F}_0(0, Q, J) = \alpha \exp\left[-\frac{\sin^2(Q/2)}{\sigma_Q^2}\right] J^2 \exp\left(-\frac{J^2}{\sigma_J^2}\right), \tag{42}$$

where (Q, J) are the action-angle variables of the external potential and the constant $\alpha > 0$ is determined by the mass of the gas. For $\sigma_Q \ll 1$ the gas particles are concentrated near the pericenters $Q = 0$ of their orbits, and their action variables are distributed about $J = \sigma_J$, see Fig. 1. The factor J^2 makes the DF smooth at the circular orbit $J = 0$, where the angle variable is not defined. In the absence of self-gravity, the mixing of this cloud is the test problem of section V A, and in section VI we analyze how its evolution changes when the mass of the gas is increased. We truncate the DF at $J_{max} = 6\sigma_J$, where it has decayed to about 2×10^{-14} of its maximum. Unless otherwise stated, $\sigma_Q = \sigma_J = 0.1$ and $N_J \times N_Q = 400 \times 25$. For the isochrone and the Kepler potentials the inversion of the map $(r, p_r) \mapsto (Q, J)$ only requires the solution of an equation of Kepler's type (see section II D), which we solve with Newton's method. In the Kepler case, for instance,

$$r = a(1 - e\cos\eta), \qquad p_r = \frac{\sqrt{a}\, e \sin\eta}{r}, \qquad Q = \eta - e\sin\eta, \tag{43}$$

with $a := (J + L_0)^2$ and $e := (1 - L_0^2/a)^{1/2}$. When the self-gravity of the gas is taken into account, the observables are computed with the action-angle variables of the instantaneous potential, see section II E.

V. CODE VALIDATION

In this section we validate the numerical methods described in the previous section. We start with the mixing of the gas cloud (42) in the absence of self-gravity, for which the exact solution is known. Next, we include the self-gravity of the cloud and show that the functions $h_n(t)$ need to be computed with the action-angle variables which belong to the total potential. Finally, we test the computation of the gravitational field in the collapse of a cold homogeneous sphere, and we report the conservation of the total energy in the simulations of section VI.

A. Mixing in the absence of self-gravity

When the last term in the second equation in (35) is omitted, the computational particles move independently from each other in the external potential, and the functions $h_n(t)$ are given by Eq. (31). This provides a test for the time integration, for the representation of the DF by the lattice (41) and for the computation of the observables. We consider the reduced model with $L_0 = 2$ in the isochrone potential and the gas cloud (42) with $\sigma_Q = \sigma_J = 0.1$ (see Fig. 1), and we choose the weight function $B(J) = J^2 \exp(-J^2/\sigma_J^2)$, that is, Eq. (40) with $J_1 = 0$ and $\sigma_1 = \sigma_J$.

The dependency of the DF (42) on the angle variable has been chosen such that it is smooth and 2π-periodic and such that its Fourier coefficients (30) are known explicitly. In fact, the identity $\sin^2(Q/2) = (1 - \cos Q)/2$ yields

$$\exp\left[-\frac{\sin^2(Q/2)}{\sigma_Q^2}\right] = e^{-z}e^{z\cos Q} = e^{-z}\sum_{n\in\mathbb{Z}} I_n(z)e^{inQ}, \qquad z := \frac{1}{2\sigma_Q^2}, \tag{44}$$

where I_n denotes the modified Bessel function of the first kind of order n, which satisfies $I_{-n} = I_n$. The second equality in Eq. (44) is the generating function of these functions, see Eq. (10.35.2) in Ref. [55]; equivalently, $I_n(z)$ is the n-th Fourier coefficient of the function $e^{z\cos Q}$ (Eq. (10.32.3) in Ref. [55]). Consequently, the Fourier coefficients of the DF (42) with respect to the angle variable are

$$\hat{F}_n(J) = \alpha\, e^{-z} I_n(z)\, J^2 \exp\left(-\frac{J^2}{\sigma_J^2}\right). \tag{45}$$

For $\sigma_Q = 0.1$ one has $z = 50$, and in this case $I_n(z)/I_0(z)$ differs from $e^{-n^2/(2z)}$ by less than one percent for $|n| \le 16$. Hence, the narrower the pulse is in the angle variable, the more harmonics it contains: here the amplitudes of those with $|n| \le 2\sqrt{z} \approx 14$ are larger than $e^{-2} \approx 0.14$ times the one of the harmonic $n = 0$. It follows from Eqs. (31,45) that $|h_n(0)|/h_0 = I_n(z)/I_0(z)$, which is equal to 0.990, 0.960, 0.913 and 0.851 for $n = 1, 2, 3, 4$, and that $h_n(t)$ is given by an integral over J, which we compute with the trapezoidal rule on 2×10^5 subintervals of $0 \le J \le 6\sigma_J$.

N_J	N_Q	Δt	e_1	e_2	e_3	e_4
50	64	0.1	4.1×10^{-6}	1.6×10^{-4}	0.15	0.72
100	64	0.1	3.3×10^{-8}	1.2×10^{-7}	5.1×10^{-7}	3.9×10^{-6}
200	64	0.1	5.3×10^{-9}	5.4×10^{-9}	5.9×10^{-9}	5.5×10^{-9}
400	64	0.1	5.3×10^{-9}	5.5×10^{-9}	6.0×10^{-9}	5.5×10^{-9}
400	16	0.1	9.2×10^{-3}	3.7×10^{-2}	8.2×10^{-2}	0.14
400	32	0.1	1.7×10^{-5}	7.3×10^{-5}	1.8×10^{-4}	3.7×10^{-4}
400	64	0.2	8.6×10^{-8}	8.8×10^{-8}	9.6×10^{-8}	8.8×10^{-8}
400	64	0.05	3.3×10^{-10}	3.4×10^{-10}	3.7×10^{-10}	3.4×10^{-10}
400	64	0.025	2.1×10^{-11}	2.0×10^{-11}	2.2×10^{-11}	2.7×10^{-11}
400	64	exact	6.6×10^{-12}	1.3×10^{-11}	2.0×10^{-11}	2.7×10^{-11}

TABLE I: Error e_n defined in Eq. (46) for the mixing of the initial datum (42) in the absence of self-gravity, for different lattices and time steps. In the last row the Yoshida integrator is replaced with the exact advance of the angle variables.

Table I shows the error

$$e_n := \frac{1}{h_0} \max_{0\le t\le 2000} \left|h_n(t) - h_n^{exact}(t)\right| \tag{46}$$

for different lattices and time steps. For $N_J \times N_Q = 400 \times 64$ and $\Delta t = 0.1$ it does not exceed 6×10^{-9} for $n \le 4$. In this case the error is dominated by the time integration. In fact, when the Yoshida integrator is replaced with the

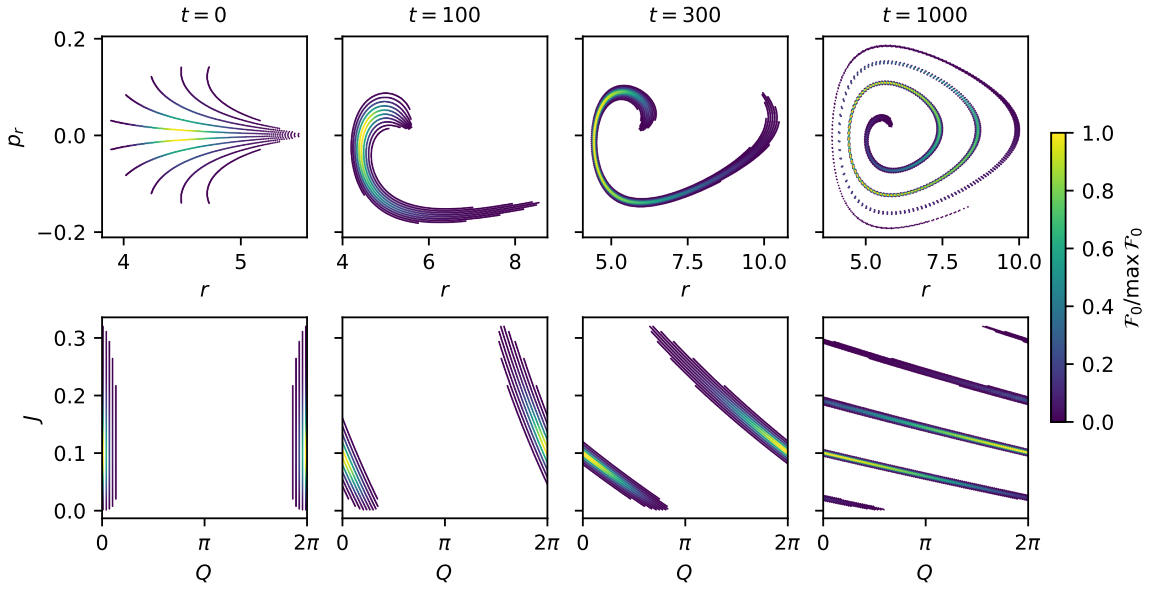


FIG. 1: Mixing of the initial datum (42) in the isochrone potential when self-gravity is neglected. Shown are the computational particles in the (r, p_r)-plane (upper panels) and in the plane of the action-angle variables (lower panels) at four different times, for a lattice with $N_J \times N_Q = 800 \times 64$ points. The color indicates the mass of each particle relative to the largest one, and only particles whose mass exceeds 10^{-3} of the latter are shown.

exact advance $Q_j \mapsto Q_j + \Omega(J_j)\Delta t$ of the angle variables of the particles, the error drops to less than 3×10^{-11}, and the difference with respect to this solution decreases by the factors 16.0, 16.0 and 15.4 each time Δt is halved between 0.2 and 0.025 (for $n = 1$), as expected for a fourth-order method.

The remaining rows of Table I show the effect of the lattice. Since the sum (39) is a midpoint rule for an integral whose integrand is smooth and, with respect to Q, periodic, the error decreases rapidly when N_J or N_Q increase, until it reaches the one of the time integration.

With respect to the angle variable the error of the lattice can be computed explicitly. On a lattice with N_Q points per row the harmonics $n + mN_Q$, $m \in \mathbb{Z}$, cannot be distinguished from the harmonic n, such that the sum over the points Q_b yields $\sum_m (-1)^m \hat{F}_{n+mN_Q}$ instead of $\hat{F}_n$, where the signs are due to the fact that the Q_b are the midpoints of the subintervals. Since the transport does not mix different harmonics, this factor is independent of time. Taking into account that the masses of the computational particles are normalized such that their sum is equal to M_{gas}, Eq. (45) gives

$$e_n = \left| \frac{\sum_m (-1)^m I_{n+mN_Q}(z)}{\sum_m (-1)^m I_{mN_Q}(z)} - \frac{I_n(z)}{I_0(z)} \right|, \tag{47}$$

where the sums extend over all $m \in \mathbb{Z}$. This expression reproduces the values in the rows with $N_Q = 16$ and $N_Q = 32$ of Table I to nine significant digits, and for $N_Q = 64$ it is smaller than 5×10^{-15}. Hence, the number of points per row has to be a few times larger than the number $2\sqrt{z}$ of harmonics contained in the DF.

With respect to the action variable the situation is different, since for a given lattice the sum is only reliable over a finite time span. The reason is that the phases $n\Omega(J_a)t$ of two neighboring rows differ by $n|\Omega'(J)|\Delta J\, t$, with ΔJ the spacing between the rows. When this difference reaches 2π the rows are in phase again, such that h_n exhibits a spurious recurrence [44] at the time

$$t_{rec} = \frac{2\pi}{n|\Omega'(J)|\Delta J}. \tag{48}$$

In the present example, $\Delta J = 0.6/N_J$ and $|\Omega'(J)| \leq 3/c(L_0)^4 \approx 0.088$, which gives $t_{rec} \approx 120\, N_J/n$. For $N_J = 50$ this is 1980 and 1480 for $n = 3$ and $n = 4$, respectively, which explains the large values of e_3 and e_4 in the first row of the table. For the lattices which are used in the next subsection and in section VI the recurrence time for $n = 1$ exceeds the duration of the simulations: computed with the frequencies of the external potential, it is 4.7×10^4 for the one of the next subsection and at least 2.2×10^4 for those of section VI. For the higher harmonics it is shorter by the factor $1/n$.

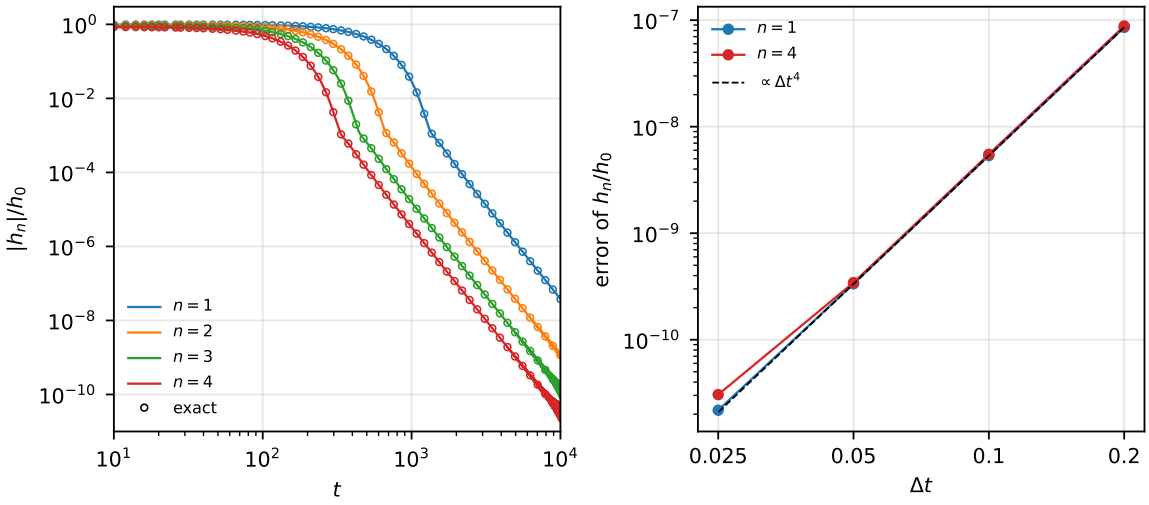


FIG. 2: Left panel: the functions $|h_n(t)|/h_0$, $n = 1, 2, 3, 4$, for the mixing of the initial datum (42) in the absence of self-gravity, obtained from the PIC code with $N_J \times N_Q = 800 \times 64$ and $\Delta t = 0.1$ (solid lines) and from the exact solution (31) (circles). Right panel: error of the time integration, measured through the difference with the solution in which the angle variables are advanced exactly, as a function of the time step. The dashed line is proportional to Δt^4.

Figure 2 shows the result of a simulation with $N_J \times N_Q = 800 \times 64$ and $\Delta t = 0.1$ which extends to $t = 10^4$. During this time $|h_1|$ and $|h_4|$ decay by seven and ten orders of magnitude, respectively, and the numerical solution reproduces the exact one over the whole range: for $n \leq 4$ the difference does not exceed $6 \times 10^{-9} h_0$ at any time, and it is smaller than $2 \times 10^{-10} h_0$ for $t \geq 5000$. At late times $|h_n(t)|$ is proportional to t^{-5} (a fit over $5000 \leq t \leq 10^4$ gives the exponents -5.03 and -5.01 for $n = 1$ and $n = 2$). This is the analogue of the algebraic tail (33) for the endpoint $J = 0$ of the support, at which the function $\hat{F}_n B$ vanishes like J^4. In the same simulation h_0, which is constant for the exact solution, varies by 1.1×10^{-9}, and the total energy by 7×10^{-11}, in relative terms.

B. Action-angle variables of the total potential

As emphasized in section II E, the functions $h_n(t)$ with $n \neq 0$ measure the departure of the gas configuration from a steady state only if they are computed with the action-angle variables which belong to the total potential. To illustrate this point we have repeated the simulation of the previous subsection including the self-gravity of the gas, for $M_{gas}/M = 10^{-4}$, 10^{-3} and 10^{-2}, with $N_J \times N_Q = 400 \times 25$, $\Delta r = 0.1$, $\Delta t = 0.1$ and up to $t = 2 \times 10^4$, using the weight function (40) with $J_1 = \sigma_1 = 0.1$. The cloud is far from a steady state, and the gas undergoes mixing while its potential Φ_{gas} readjusts: for $M_{gas}/M = 10^{-3}$ it changes by up to 40 percent of its largest value between $t = 0$ and $t = 2000$, and afterwards it is stationary up to fluctuations whose root mean square is smaller than 10^{-3} of this value.

Figure 3(a) shows the function $|h_1(t)|/h_0$ which is obtained with the action-angle variables of the external potential. When self-gravity is neglected it decays according to the law t^{-5} found in the previous subsection, until it reaches the level of the numerical errors, which is of the order of 10^{-9}, at $t \approx 10^4$. In contrast to this, when self-gravity is taken into account it settles down to a plateau: the magnitude of the time average of h_1/h_0 over the interval $2000 \leq t \leq 2 \times 10^4$ is 2.2×10^{-4}, 2.2×10^{-3} and 1.5×10^{-2} for the three values of M_{gas}/M, whereas it is 9×10^{-9} in the absence of self-gravity. For $M_{gas}/M = 10^{-3}$ the same value, 2.16×10^{-3}, is obtained with $N_J \times N_Q = 800 \times 25$ and 400×50. Such a plateau could be mistaken for a collective mode which is not damped.

However, the plateau disappears when the same particle data are analyzed with the action-angle variables of the potential $\Phi_{ext} + \langle \Phi_{gas} \rangle$, where $\langle \Phi_{gas} \rangle$ denotes the time average of the potential of the gas after it has become stationary. This is shown in Fig. 3(b) for $M_{gas}/M = 10^{-3}$: with these variables h_1/h_0 fluctuates about zero instead of about the value 2.16×10^{-3}, and the magnitude of its time average over $2000 \leq t \leq 2 \times 10^4$ is reduced to 5.4×10^{-7}, which is 4000 times smaller. The action-angle variables of the instantaneous potential, which we use for the initial data of section VI, give the same result: with them the magnitude of the time average is 1.5×10^{-6}. Therefore, the plateau is due to the choice of the coordinates and not to the dynamics. The reason is that the action variable of the external potential is not conserved along the trajectories of the gas particles in the total potential. As shown in Fig. 3(c), for a given particle it oscillates with the period of the radial motion; the relative difference between its largest and its

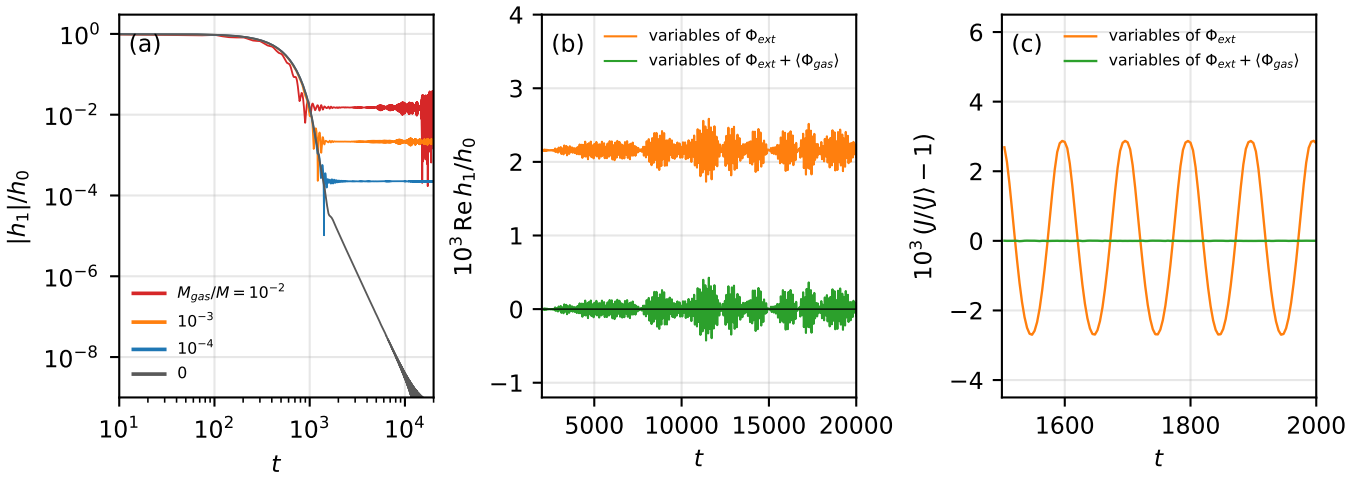


FIG. 3: Dependency of the function h_1 on the action-angle variables which are used to compute it. (a) $|h_1|/h_0$ for the mixing of the initial datum (42) including self-gravity, computed with the action-angle variables of the external potential, for different masses of the gas. (b) Real part of h_1/h_0 for $M_{gas}/M = 10^{-3}$, computed with the action-angle variables of the external potential and with those of the potential $\Phi_{ext} + \langle\Phi_{gas}\rangle$, where $\langle\Phi_{gas}\rangle$ is the average of Φ_{gas} over $t \geq 2000$. (c) Relative variation of the action variable of a single computational particle for the same mass, for both sets of variables; here $\langle\Phi_{gas}\rangle$ is the average over $1500 \leq t \leq 2000$, and $\langle J\rangle$ denotes the mean value over this interval.

smallest value is 5.6×10^{-3} for the particle shown, which has $J = 0.10$, and 4.4×10^{-3} for a particle with $J = 0.30$. In contrast to this, the action variable which belongs to the total potential is constant to within a relative variation of 2×10^{-5}. Consequently, a DF which depends only on the action variable of the total potential is not independent of the angle variable of the external one, the difference between both sets of variables being of the first order in M_{gas}/M.

What remains after subtracting the time average is a fluctuation which is the same for both sets of variables, see Fig. 3(b). Its root mean square is $2.0 \times 10^{-4} h_0$ in the simulation with 400×25 particles, and $1.1 \times 10^{-4} h_0$ and $9.6 \times 10^{-5} h_0$ in those with 800×25 and 400×50 particles, respectively. For $M_{gas}/M = 10^{-2}$ it grows in time and is visible in Fig. 3(a) for $t > 10^4$; in this case its root mean square is $9.1 \times 10^{-3} h_0$ with 400×25 particles and $2.2 \times 10^{-3} h_0$ with 800×25 particles. Hence, these fluctuations are caused by the finite number of particles.

C. Self-gravity and conservation of the energy

We have tested the computation of the gravitational field in a situation which is dominated by self-gravity, namely the collapse of a cold homogeneous sphere of mass M_{gas} and radius R in the absence of an external potential. Since in this case all the particles have $L = 0$, this test has been performed with an implementation of the same algorithm for a general distribution of the angular momenta. Before the shells cross each other, the shell with initial radius r_0 follows the cycloid $r = r_0 \cos^2\theta$, $t = (\theta + \sin\theta\cos\theta)/\sqrt{2GM_{gas}/R^3}$. For $\Delta r = 0.01R$, the median of the relative error of the radii of the shells with $0.1 < r_0/R < 0.9$ at the time $t = 0.8\sqrt{R^3/(GM_{gas})}$ is 1.8×10^{-5} for $N = 200$ and 4.3×10^{-6} for $N = 400$ shells, which corresponds to second-order convergence in $1/N$.

In the simulations of the gas cloud in section VI, which use $\Delta t = 0.05$ and extend to $t = 10^4$ (2×10^4 for $\sigma_J = 0.05$), the largest relative change of the total energy (38) is 1.7×10^{-4} for $M_{gas}/M = 0.3$, it does not exceed 7.2×10^{-5} for the remaining simulations, and its median is 1.0×10^{-5}. [(Erik) A test of the dependency on the time step for the cloud (for instance $M_{gas}/M = 0.01$ and 0.1 with $\Delta t = 0.1$ and 0.025) is to be added; it requires new simulations.]

VI. INITIAL DATA FAR FROM EQUILIBRIUM

In this section we consider the gas cloud (42), which is concentrated about one orbital phase and is far from any steady state, and we analyze how its evolution changes when its mass is increased. We start with the simulations and with a parameter μ which compares the amplitude of the potential generated by the cloud with the spread of its orbital frequencies. Next, we describe the evolution of the cloud for increasing values of μ: for small μ the cloud mixes; for larger μ a compact part of it survives as a dense core at the circular orbit, whose radial oscillation is damped; and

above a critical value of μ this oscillation persists, such that the potential does not relax. Then, we characterize the state of the gas at late times, and finally we analyze the dependence of these results on the widths of the cloud and on the external potential.

A. Simulations and the parameter μ

The reference family of simulations consists of the cloud (42) with $\sigma_Q = \sigma_J = 0.1$ and $L_0 = 2$ in the isochrone potential, with the masses $M_{gas}/M = 10^{-3}$, 0.01, 0.015, 0.02, 0.025, 0.03, 0.04, 0.05, 0.07, 0.1 and 0.3, together with a simulation without self-gravity. In order to analyze the dependence on the shape of the cloud, we vary its width in angle, $\sigma_Q = 0.2$ and 0.4 with $M_{gas}/M = 0.01$, 0.02, 0.04 and 0.08, and its width in action, $\sigma_J = 0.05$ with $M_{gas}/M = 0.005$, 0.01 and 0.02 and $\sigma_J = 0.2$ with $M_{gas}/M = 0.02$, 0.04 and 0.08. Finally, we consider the cloud with $\sigma_Q = \sigma_J = 0.1$ in the Kepler potential, with $M_{gas}/M = 0.01$, 0.02, 0.04 and 0.08. This is the external potential of the nonlinear phase mixing result of Ref. [35], although the reduced model and the support of the DF, which extends to the circular orbit, differ from the assumptions made there. The simulations use the lattice (41) with $N_J \times N_Q = 400 \times 25$, except for $\sigma_Q = 0.2$ ($N_Q = 50$) and $\sigma_J = 0.2$ ($N_J = 800$), and the time step $\Delta t = 0.05$, and they extend to $t = 10^4$ (2×10^4 for $\sigma_J = 0.05$). In order to test the convergence, we have repeated the simulations of the reference family with $M_{gas}/M = 0.01$, 0.02 and 0.1 with $N_Q = 100$, and the one with $M_{gas}/M = 0.01$ with $N_J = 800$. In all the simulations the gas particles remain bound, and the observables are computed with the action-angle variables of the instantaneous potential and the weight function $B = 1$, see sections II E and IV A.

In the absence of self-gravity the cloud mixes on a time scale which is set by the spread of its orbital frequencies. We characterize this spread by the interval $[\underline{J}, \overline{J}]$ in which the factor $J^2 \exp(-J^2/\sigma_J^2)$ of the DF (42) exceeds one percent of its maximum, by its width $\Delta J := \overline{J} - \underline{J}$, and by the width $\Delta\Omega := \Omega(\underline{J}) - \Omega(\overline{J})$ of the corresponding frequency band of the external potential. After the time

$$\tau_1 := \frac{2\pi}{\Delta\Omega} \tag{49}$$

the orbits at the two ends of this interval have drifted apart by one revolution, such that the first harmonic of the cloud has been spread over the orbit. For $\sigma_Q = \sigma_J = 0.1$ in the isochrone potential one has $\Delta J = 0.270$, $[\Omega(\overline{J}), \Omega(\underline{J})] = [0.0513, 0.0705]$ and $\tau_1 = 327$, for $\sigma_J = 0.05$ and 0.2 one has $\tau_1 = 589$ and 198, respectively, and in the Kepler potential $\tau_1 = 161$. Hence, the simulations extend over at least 30 units of τ_1.

The self-gravity of the cloud couples the motions of its particles and acts against the mixing. In order to compare both effects, consider the motion of the gas particles relative to the center of the cloud. In the absence of self-gravity, the angle variable of a particle relative to the one of the center drifts with a velocity which varies by $\Delta\Omega$ across the support of the cloud. Approximating the derivative of the frequency by the constant $|\Omega'| = \Delta\Omega/\Delta J$, the energy which is associated with this differential drift, $|\Omega'|(J - J_c)^2/2$ with J_c the action variable of the center, is of the order of $\Delta\Omega\,\Delta J$. On the other hand, the strength of the self-gravity is measured by the amplitude A of the variation of the potential which is generated by the cloud. This suggests the parameter

$$\mu := \left(\frac{A}{\Delta\Omega\,\Delta J}\right)^{1/2} = \frac{\omega_b}{\Delta\Omega}, \qquad \omega_b := (|\Omega'|A)^{1/2}, \tag{50}$$

whose square compares these two energies, and one expects the self-gravity of the cloud to modify the mixing when μ is of order one. The frequency ω_b is the frequency of the small oscillations of a pendulum with the Hamiltonian $|\Omega'|(J - J_c)^2/2 - A\cos\vartheta$, which describes the trapping of particles by a wave in a plasma [56]; in terms of μ, $2\pi/\omega_b = \tau_1/\mu$. This argument is a heuristic one: it does not describe how the potential of the cloud depends on the positions of its particles, and it does not predict the critical value of μ. We measure A as half of the largest variation of $\Phi_{gas}(t, r)$ at fixed r during the first radial period $0 \le t \le 2\pi/\Omega(\overline{J})$, maximized over $3 \le r \le 15$. For small masses A is determined by the initial datum: in the reference family with $M_{gas}/M \le 0.025$ it differs by less than 5 percent from $0.0496\,M_{gas}/M$, where 0.0496 is the amplitude, per unit mass, of the potential which the cloud generates when it moves in the external potential alone. Hence, μ grows like $(M_{gas}/M)^{1/2}$. For $M_{gas}/M = 10^{-3}$, 0.01, 0.1 and 0.3 one has $\mu = 0.10$, 0.31, 0.89 and 1.16, respectively.

B. Evolution of the cloud

Figure 4 shows the computational particles in the (r, p_r)-plane for three clouds of the reference family. For $M_{gas}/M = 0.01$ ($\mu = 0.31$) the evolution resembles the mixing without self-gravity of Fig. 1: the cloud is sheared

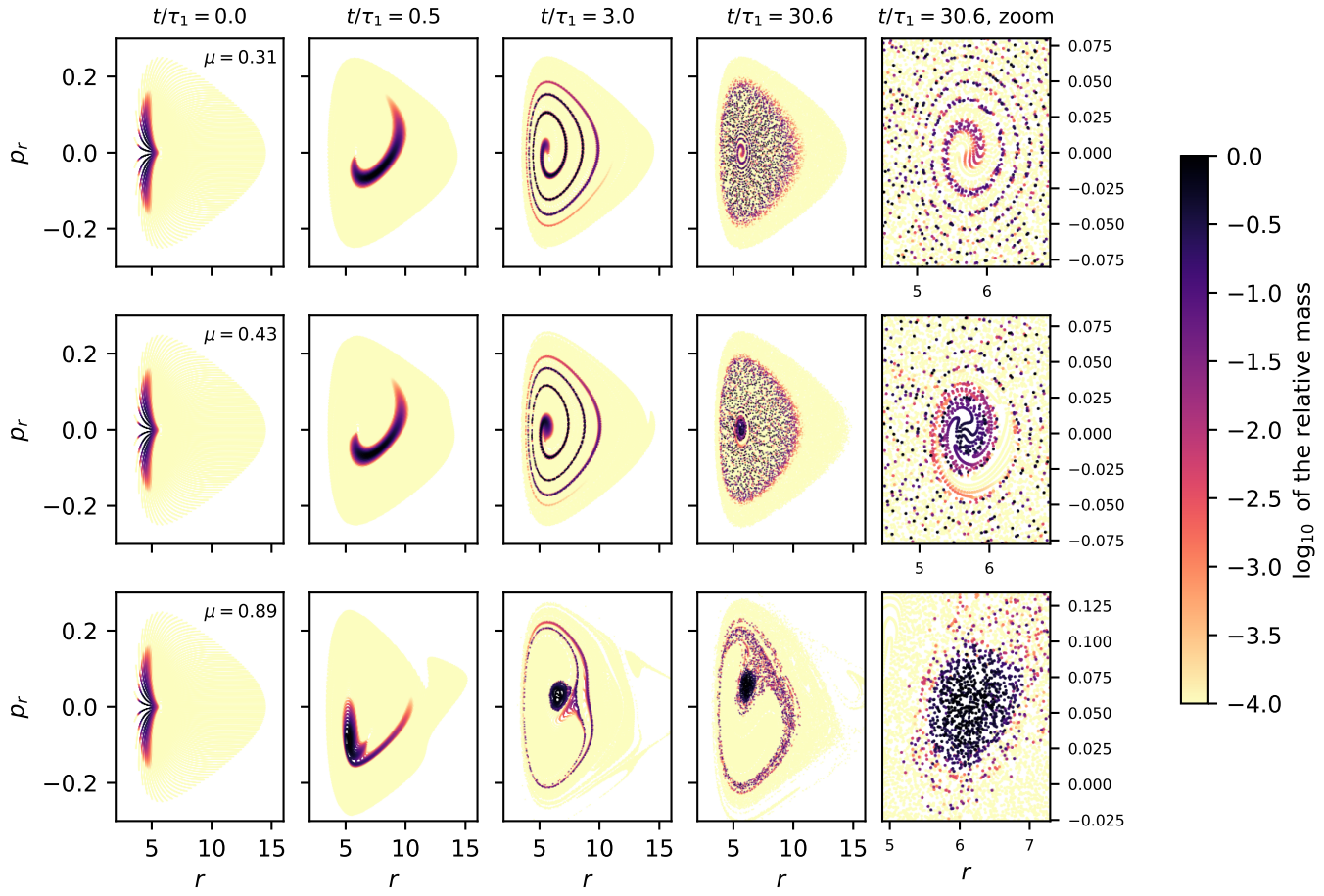


FIG. 4: The computational particles in the (r, p_r)-plane at four different times for the cloud (42) with $\sigma_Q = \sigma_J = 0.1$ in the isochrone potential, for $M_{gas}/M = 0.01$, 0.02 and 0.1 (from top to bottom; $N_J \times N_Q = 400 \times 100$). The color indicates the mass of each particle relative to the largest one. The last column shows, at the final time, the neighborhood of the circular orbit (upper two rows) and of the core (lower row). For $\mu = 0.31$ the cloud mixes, for $\mu = 0.43$ a dense core is left at the circular orbit, and for $\mu = 0.89$ the core oscillates radially until the end of the simulation.

into a spiral which winds up around the circular orbit, and at $t = 30\tau_1$ the gas fills the region of phase space which is covered by its orbits. For $M_{gas}/M = 0.02$ ($\mu = 0.43$) most of the gas evolves in the same way, but a compact part of the cloud ends up at the circular orbit, where it forms a dense core which is visible from $t = 3\tau_1$ on. For $M_{gas}/M = 0.1$ ($\mu = 0.89$) about half of the cloud contracts within the first radial periods into a core which is not at rest: for $t \geq 15\tau_1$ its center moves between $r = 4.7$ and $r = 6.7$ with the frequency 0.074, which is close to the frequency $\Omega(0) = 0.0711$ of the circular orbit in the external potential, while the root mean square distance of its particles from the center is 0.17. The rest of the gas is mixed around it.

The function $|h_1(t)|/h_0$ and the oscillation of the potential are shown in Fig. 5 for the reference family. Without self-gravity, $|h_1|$ decays like t^{-3} (a fit of the envelope over $10 \leq t/\tau_1 \leq 30$ gives the exponent -3.03), which is the analogue of the tail (33) for the endpoint $J = 0$ of the support, at which $\hat{F}_1$ vanishes like J^2. With self-gravity the decay slows down, and the level which is reached at late times increases with μ. Averaged over $10 \leq t/\tau_1 \leq 20$, $|h_1|/h_0$ is 1.1×10^{-4}, 2.4×10^{-3} and 1.2×10^{-2} for $\mu = 0.10$, 0.31 and 0.43 (the last two values for $N_Q = 100$), compared to 7.0×10^{-5} without self-gravity, and it lies between 0.09 and 0.85 for $\mu \geq 0.48$.

The potential which is generated by the gas behaves in the same way, see Fig. 5(b). Without self-gravity, the potential which the gas would generate relaxes to its final profile through the mixing alone: its deviation (27) from the average over the second half of the simulation falls below 10^{-3} of the largest value of $|\langle\Phi_{gas}\rangle|$ at $t = 2.9\tau_1$. With self-gravity and $\mu \leq 0.43$ the potential still relaxes, but more slowly: the same level is reached at $t = 3.3\tau_1$, $4.8\tau_1$, $5.5\tau_1$ and $6.2\tau_1$ for $\mu = 0.10$, 0.31, 0.37 and 0.43, respectively. Over $10 \leq t/\tau_1 \leq 20$ the deviation lies between 7×10^{-5} and 3×10^{-4}, compared to 4×10^{-5} without self-gravity, and it is reduced when the number of particles is increased (dotted lines). For $\mu \geq 0.48$, in contrast, the deviation never falls below 10^{-3}: over $10 \leq t/\tau_1 \leq 20$ it lies between 1.6×10^{-3}

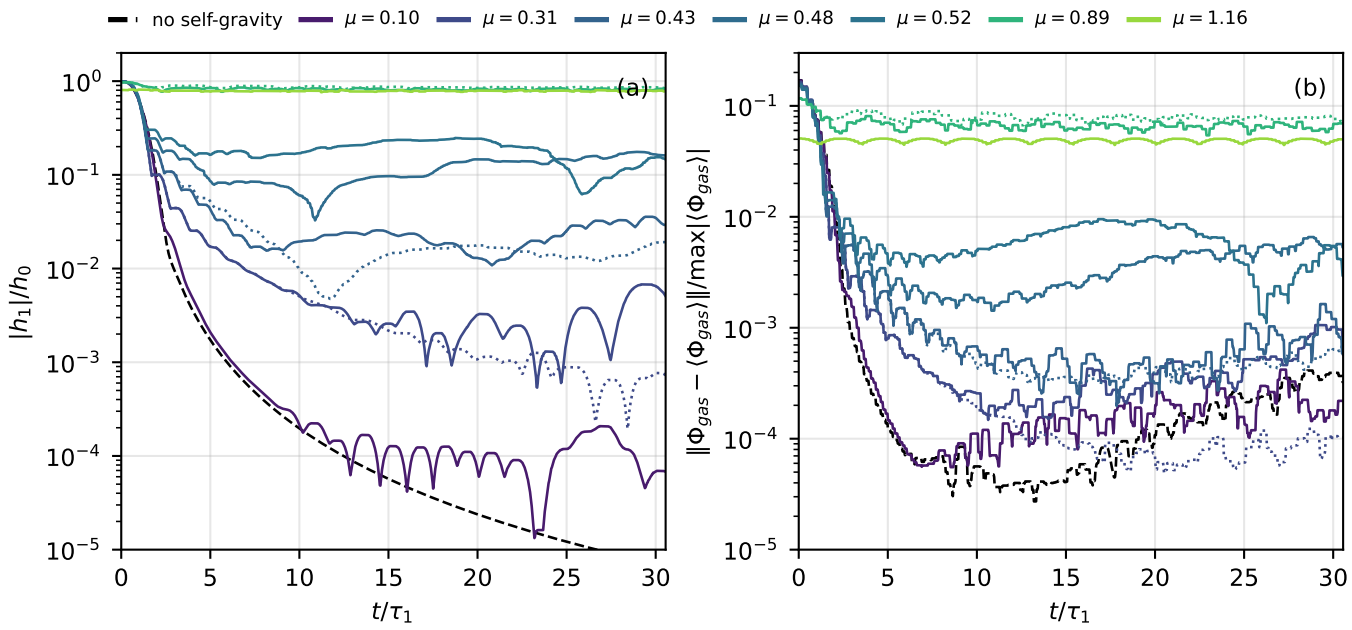


FIG. 5: Evolution of the cloud (42) with $\sigma_Q = \sigma_J = 0.1$ in the isochrone potential for different masses, labeled by the value of μ. (a) Envelope of $|h_1|/h_0$ (largest value in a moving window of width 100), computed with the action-angle variables of the instantaneous potential, for $N_J \times N_Q = 400 \times 25$ (solid lines) and 400×100 (dotted lines). (b) Deviation of the potential generated by the gas from its average over the second half of the simulation, measured with the norm (28) over $3 \le r \le 15$ in units of the largest value of $|\langle\Phi_{gas}\rangle|$ (envelope as in (a)). The potential is computed from the radii of the particles; without self-gravity (dashed line) it is the potential which the gas would generate if it acted on itself.

and 4.6×10^{-2}, since the potential oscillates with the core. Therefore, for $\mu \le 0.43$ the gravitational field generated by the gas relaxes to a stationary field while the DF continues to mix, which is the analogue of Landau damping, whereas for $\mu \ge 0.48$ the field does not relax. (Without self-gravity the deviation increases again for $t \gtrsim 15\tau_1$, which we attribute to the spurious recurrence (48) of the higher harmonics of the potential on the lattice; for $n = 10$ the recurrence time is about $4800 \approx 15\tau_1$.)

C. The state at late times: dense core and its oscillation

In the absence of self-gravity the gas relaxes to the angle average $\hat{F}_0(J)$ of the initial DF, see section III. For the cloud (42), Eq. (45) gives $\hat{F}_0(J) = \alpha\, e^{-z} I_0(z) J^2 \exp(-J^2/\sigma_J^2)$, whose maximum $\hat{F}_{max} = \alpha\, e^{-z} I_0(z) \sigma_J^2/e$ is attained at $J = \sigma_J$. This is the largest value of the DF which can be reached anywhere by mixing the cloud at fixed action. It is much smaller than the maximum $F_{max} = \alpha\sigma_J^2/e$ of the initial DF itself: $F_{max}/\hat{F}_{max} = 17.7$, 8.8 and 4.2 for $\sigma_Q = 0.1$, 0.2 and 0.4, respectively. In order to characterize the state which is reached with self-gravity, we compute the average of the DF over the angle variable of the instantaneous potential, as a function of the action variable of the same potential, from the mass of the computational particles in a set of intervals of J, and we average the result over $t \ge 20\tau_1$.

The result is shown in Fig. 6(a). Without self-gravity the angle average of the DF does not change, as it should. For $\mu = 0.31$ it remains a smooth function which lies below $\hat{F}_{max}$ everywhere (its largest value is $0.67\,\hat{F}_{max}$), but it differs from $\hat{F}_0$: the region near the circular orbit, where the initial DF vanishes like J^2, has been partially filled. The fraction of the mass of the gas with $J < 0.2\sigma_J$ is 6.1 percent, compared to 0.5 percent without self-gravity. Hence, although the gas mixes, the state which it reaches is not the angle average of the initial DF. The reason is that during the first few units of τ_1 the potential of the cloud oscillates with the orbital period of its particles, such that their action variables are not conserved; the fraction quoted above grows from 1.0 to 5.8 percent between $t = 0$ and $t = 3\tau_1$ and changes by less than 0.3 percent afterwards. This fraction increases continuously with μ, see Table II, and it is the same for the three lattices which we have used for $\mu = 0.31$ (6.0, 6.1 and 6.1 percent).

For $\mu \ge 0.37$ a peak develops at the circular orbit which exceeds $\hat{F}_{max}$. In the interval $J < 0.025\sigma_J$ the angle average of the DF is $1.3\,\hat{F}_{max}$ for $\mu = 0.37$, about $4\,\hat{F}_{max}$ for $\mu = 0.43$ (4.3 and 3.6 with $N_Q = 25$ and 100), and 8.3

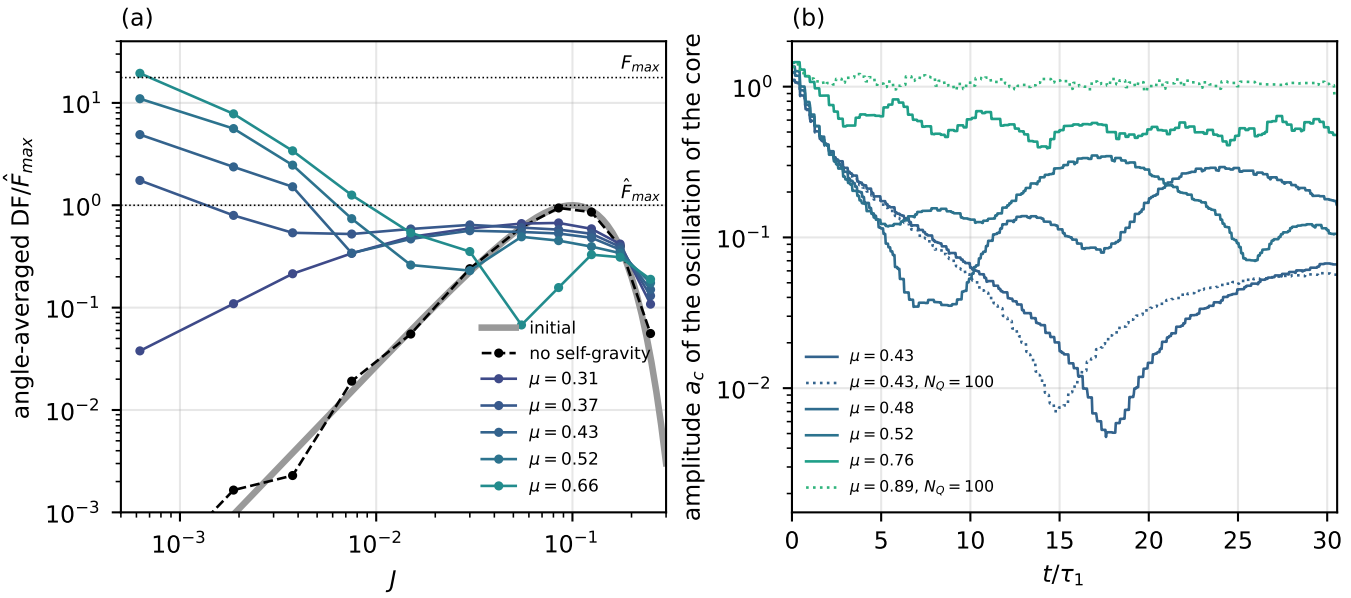


FIG. 6: State of the cloud (42) with $\sigma_Q = \sigma_J = 0.1$ in the isochrone potential at late times. (a) Angle average of the DF as a function of the action variable of the instantaneous potential, averaged over $t \geq 20\tau_1$, in units of the maximum $\hat{F}_{max}$ of the angle average of the initial DF (thick gray line). The dotted lines indicate $\hat{F}_{max}$ and the maximum F_{max} of the initial DF. The results refer to $N_Q = 100$ for $\mu = 0.31$ and 0.43 and to $N_Q = 25$ otherwise. (b) Amplitude a_c of the radial oscillation of the core (see the text), for $N_Q = 25$ (solid lines) and $N_Q = 100$ (dotted lines).

and $13.6\,\hat{F}_{max}$ for $\mu = 0.52$ and 0.66; in the innermost interval of Fig. 6(a) it reaches F_{max} for $\mu = 0.66$, within the sampling error. Values above $\hat{F}_{max}$ cannot be obtained by mixing the cloud at fixed action. They show that part of the dense cloud has been transported to the circular orbit before it was mixed. A distribution which is centered at the circular orbit is not dispersed by the differential rotation of the orbits about it, which only winds up its internal structure (see the last column of Fig. 4). We refer to this part of the gas as the core.

M_{gas}/M	μ	$\lvert h_1\rvert/h_0$	mass with $J < 0.2\sigma_J$	m_c	m_c/m_{mix}	a_c for t/τ_1 in $[1,2]$	$[20,30]$
0	0	7.0×10^{-5}	0.005	0.051	1.2	–	–
0.001	0.10	1.1×10^{-4}	0.007	0.048	1.2	–	–
0.01	0.31	2.4×10^{-3}	0.061	0.031	0.7	–	–
0.015	0.37	8.5×10^{-3}	0.108	0.040	1.0	–	–
0.02	0.43	1.2×10^{-2}	0.160	0.103	2.5	0.51	0.06
0.025	0.48	0.091	0.225	0.173	4.2	0.55	0.27
0.03	0.52	0.22	0.276	0.232	5.6	0.55	0.24
0.04	0.60	0.32	0.377	0.337	8.1	0.57	0.29
0.05	0.66	0.19	0.450	0.376	9.1	0.49	0.24
0.07	0.76	0.58	0.557	0.479	11.5	0.93	0.58
0.1	0.89	0.85	0.290	0.516	12.4	1.00	0.99
0.3	1.16	0.69	0.749	0.471	11.3	0.59	0.63

TABLE II: State at late times of the cloud (42) with $\sigma_Q = \sigma_J = 0.1$ in the isochrone potential: mean value of $\lvert h_1\rvert/h_0$ over $10 \leq t/\tau_1 \leq 20$; fraction of the mass of the gas whose action variable with respect to the instantaneous potential is smaller than $0.2\sigma_J$, and fraction m_c inside the ellipse (51) about the maximum of the density, both averaged over $t \geq 20\tau_1$; and half of the difference between the largest and the smallest radius of the center of the core in the first and in the last time interval. For a gas which is mixed at fixed action $m_c \leq m_{mix} = 0.042$, and no amplitude is given when m_c is compatible with this bound. For $\mu = 0.89$ the core does not lie at the circular orbit, which explains the smaller value in the fourth column. The rows with $\mu = 0.31$, 0.43 and 0.89 refer to $N_Q = 100$.

In order to follow the core also when it is not at rest, we determine at each time the maximum (r_*, p_*) of the density

of the gas in the (r, p_r)-plane and consider the ellipse

$$\frac{(r-r_*)^2}{a^2} + \frac{(p_r-p_*)^2}{\kappa^2 a^2} < 1, \qquad a := \left(\frac{2J_*}{\kappa}\right)^{1/2}, \qquad J_* := 0.05\,\sigma_J, \tag{51}$$

where $\kappa := \Omega(0)$ is the frequency of the circular orbit of the external potential. In the harmonic approximation this is the set of orbits about (r_*, p_*) whose action variable is smaller than J_*; for $\sigma_J = 0.1$ in the isochrone potential $a = 0.38$. We denote by m_c the fraction of the mass of the gas inside this ellipse, averaged over $t \geq 20\tau_1$. Since the area of the ellipse is $2\pi J_*$, a gas which is mixed at fixed action satisfies $m_c \leq m_{mix}$, where $m_{mix} = 0.05 \cdot 4/(e\sqrt{\pi}) = 0.042$ is the mass fraction which corresponds to the density $\hat{F}_{max}$, independently of σ_Q and σ_J; hence, m_c/m_{mix} is the mean value of the DF in the ellipse in units of $\hat{F}_{max}$. As shown in Table II, m_c/m_{mix} lies between 0.7 and 1.2 for $\mu \leq 0.37$, which is compatible with one within the fluctuations due to the finite number of particles, and it grows from 2.5 for $\mu = 0.43$ to about 12 for $\mu \geq 0.76$. For $\mu = 0.43$ the core contains 10 percent of the mass of the gas, and for $\mu = 0.89$ at least 52 percent; for the largest masses the core is larger than the ellipse, such that m_c is a lower bound.

The oscillation of the core is described by the mean radius $r_c(t)$ of the computational particles which lie inside the ellipse at the end of the simulation. We denote by $a_c(t)$ the envelope of the deviation of r_c from its average over one radial period; it is shown in Fig. 6(b). In all the simulations with a core, a_c lies between 0.2 and 1.1 at $t = 2\tau_1$. Its subsequent behavior depends on μ. For $\mu = 0.43$ the oscillation is damped: a_c decays from 0.41 at $t = 2\tau_1$ to less than 7×10^{-3} at $t \approx 15\tau_1$ to $18\tau_1$, and a fit of an exponential function in the interval $2 \leq t/\tau_1 \leq 14$ gives the decay times $3.8\tau_1$ and $4.7\tau_1$ for $N_Q = 100$ and $N_Q = 25$, respectively. Afterwards a_c increases again, to 0.06 at the end of both simulations, which is 15 percent of its value at $t = 2\tau_1$ and less than half of the size of the core, whose root mean square radial extension is 0.15. For $0.48 \leq \mu \leq 0.66$ the first decay stops at $t \approx 4\tau_1$ to $9\tau_1$, when a_c has decreased by a factor between 3 and 12, and afterwards a_c is modulated, with maxima between 0.2 and 0.35 and minima between 0.02 and 0.2, without an overall decay until the end of the simulations. For $\mu = 0.76$ it remains above 0.39, and for $\mu \geq 0.89$ it does not decay. Therefore, the change between $\mu = 0.43$ and $\mu = 0.48$ which was found in the previous subsection separates the clouds whose core settles down at the circular orbit from those whose core keeps oscillating. The former reach a state which is stationary up to the small residual oscillation mentioned above, and which consists of a core and of mixed gas around it; the damping of the oscillation of the core is accompanied by the relaxation of the potential shown in Fig. 5(b). [(Erik) The increase of a_c for $\mu = 0.43$ after $t \approx 15\tau_1$ occurs with both values of N_Q, but the two simulations have the same number $N_J = 400$ of rows, and $15\tau_1$ is the recurrence time of the harmonic $n = 10$ on this lattice. A simulation with $N_J = 800$ is needed to decide whether this increase is physical.]

The modulation of a_c for $\mu \geq 0.48$ resembles the oscillations of the amplitude of a plasma wave which has trapped the resonant particles [56], and for Langmuir waves a threshold amplitude is known which separates the waves that are damped from those that are not [57]. Whether the same mechanism is at work here remains to be analyzed. Regarding the nature of the core, we note that for $\mu = 0.89$ the core, whose center oscillates with the amplitude 1.0, would be spread over its orbit by the differential rotation within about $10\tau_1$ if its particles moved independently from each other in the external potential. Since it remains compact over $30\tau_1$, it is held together by its own gravity. For $0.43 \leq \mu \leq 0.66$ the core lies so close to the circular orbit that the corresponding time exceeds $40\tau_1$, such that our simulations do not allow us to decide whether it is self-bound.

The state which consists of a compact core and of mixed gas around it resembles the core-halo states which are formed by collisionless relaxation in systems with long-range interactions [36]. In the scenario described in that reference, the particles which are in parametric resonance with the oscillation of the core escape to the halo, and this damps the oscillation until a quasi-stationary state is reached. In our simulations this is what happens for $\mu \leq 0.43$, whereas for larger values of μ the oscillation of the core is damped only slowly, if at all, over 30 units of τ_1.

D. Dependence on the shape of the cloud and on the external potential

Figure 7 shows the mean value of $|h_1|/h_0$ over the interval $10 \leq t/\tau_1 \leq 20$ and the ratio m_c/m_{mix} as functions of μ for all the simulations. In each family, $|h_1|/h_0$ grows by one to two orders of magnitude within a narrow interval of μ. With the exception of the family with $\sigma_Q = 0.4$, the location of this interval is the same for all the families: all the simulations with $\mu \leq 0.43$ have $|h_1|/h_0 \leq 2.1 \times 10^{-2}$, and all those with $\mu \geq 0.48$ have $|h_1|/h_0 \geq 0.09$. The ratio m_c/m_{mix} behaves in a similar way: it lies between 0.7 and 1.2 for all the simulations with $\mu \leq 0.38$, and it is larger than 4 for all those with $\mu \geq 0.48$ and $\sigma_Q \leq 0.2$. In between, where a core is present whose oscillation is damped, we find $m_c/m_{mix} = 1.4$ for $\sigma_Q = 0.2$ and $\mu = 0.42$, besides the value 2.5 for the reference family and $\mu = 0.43$; in these two simulations and in the one with $\sigma_J = 0.2$ and $\mu = 0.38$, in which the peak at the circular orbit is narrower than the ellipse, the oscillation of the core is damped with decay times between $4\tau_1$ and $11\tau_1$.

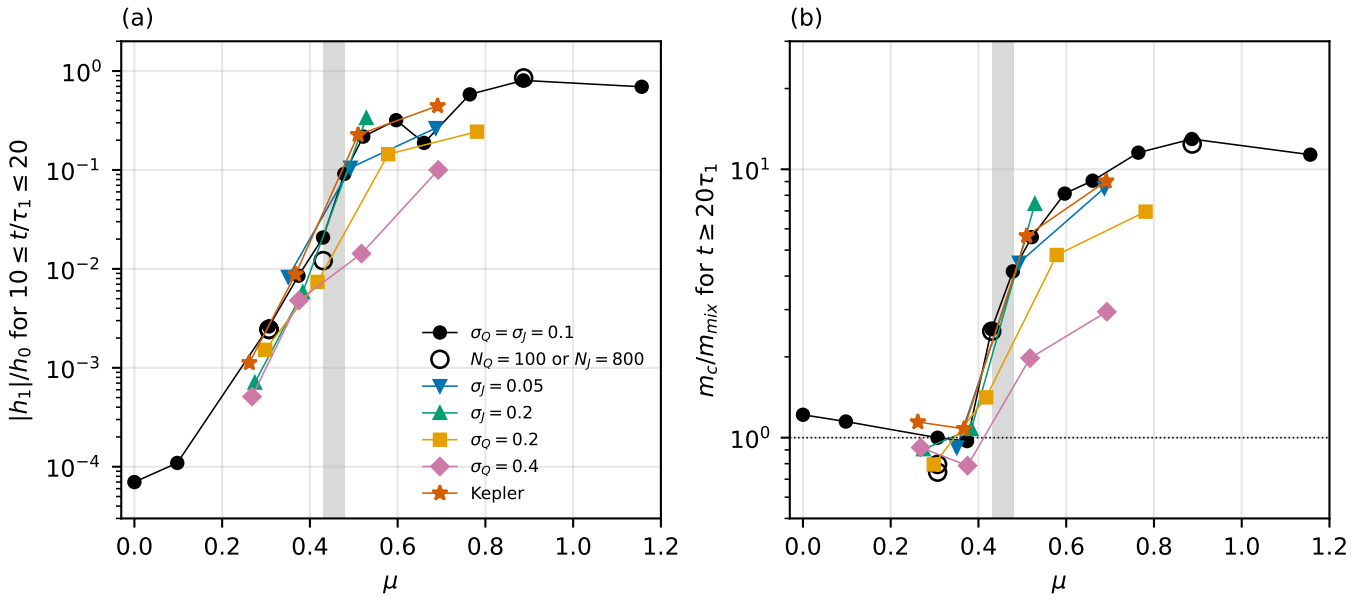


FIG. 7: All the simulations of the cloud (42) as functions of the parameter (50). (a) Mean value of $|h_1|/h_0$ over the interval $10 \le t/\tau_1 \le 20$. (b) Mass fraction m_c inside the ellipse (51) in units of the largest value m_{mix} which is compatible with a gas that is mixed at fixed action (dotted line). The families differ in the widths σ_Q and σ_J of the cloud (equal to 0.1 unless otherwise stated) and in the external potential (isochrone unless otherwise stated). The open circles refer to the simulations of the reference family with $N_Q = 100$ or $N_J = 800$. The shaded band indicates the interval $0.43 \le \mu \le 0.48$.

Table III lists, for each family, the largest value of μ for which the gas relaxes and the smallest one for which the core keeps oscillating. The intervals which are delimited by these values have the common part $0.43 \le \mu \le 0.48$ for the five families with $\sigma_Q \le 0.2$, whereas for $\sigma_Q = 0.4$ the transition lies between $\mu = 0.52$ and $\mu = 0.69$. Therefore, for clouds which are compact in angle the critical value μ_c of the parameter (50) lies between 0.43 and 0.48. This interval is determined by the reference family; for the remaining families, in which the masses are more widely spaced, the transition is only bracketed by wider intervals which contain it. A cloud which is wide in angle requires a larger value of μ. The critical value is of order one, as expected from the argument of section VI A, but this argument does not predict it. Hence, μ captures the competition between self-gravity and shear for compact clouds, but not the complete dependence on the shape of the cloud. [(Erik) A theory of μ_c, including its dependence on σ_Q, is still missing. Intermediate masses in the families other than the reference one are needed in order to test whether μ_c is the same for all of them.]

		relaxation			oscillating core		
potential	widths	μ	$\lvert h_1\rvert/h_0$	m_c/m_{mix}	μ	$\lvert h_1\rvert/h_0$	m_c/m_{mix}
isochrone	$\sigma_Q = \sigma_J = 0.1$	0.43	1.2×10^{-2}	2.5	0.48	0.091	4.2
	$\sigma_J = 0.05$	0.35	8.3×10^{-3}	0.9	0.49	0.105	4.5
	$\sigma_J = 0.2$	0.38	5.8×10^{-3}	1.1	0.53	0.33	7.4
	$\sigma_Q = 0.2$	0.42	7.4×10^{-3}	1.4	0.58	0.14	4.8
	$\sigma_Q = 0.4$	0.52	1.4×10^{-2}	2.0	0.69	0.10	2.9
Kepler	$\sigma_Q = \sigma_J = 0.1$	0.37	8.8×10^{-3}	1.1	0.51	0.23	5.6

TABLE III: Bracketing of the transition: for each family, the largest value of μ for which the gas relaxes and the smallest one for which the core keeps oscillating, together with the mean value of $|h_1|/h_0$ over $10 \le t/\tau_1 \le 20$ and the ratio m_c/m_{mix}. In the family with $\sigma_J = 0.05$ ($\sigma_Q = 0.2$, 0.4) the remaining width is $\sigma_Q = 0.1$ ($\sigma_J = 0.1$). For $\mu = 0.43$ the values refer to $N_Q = 100$; for $N_Q = 25$ they are 2.1×10^{-2} and 2.5. Recall that m_c/m_{mix} cannot exceed $F_{max}/\hat{F}_{max}$, which is 4.2 for $\sigma_Q = 0.4$.

Regarding the resolution, we note that with $N_Q = 25$ the cloud with $\sigma_Q = 0.1$ is sampled by only four columns of the lattice in the angle variable, two on each side of $Q = 0$ with the relative weights 0.67 and 0.03, such that about 600 of the 10^4 computational particles carry more than one percent of the largest weight; with $N_Q = 100$ this number is about 2000. Nevertheless, the results for both lattices agree. For $\mu = 0.31$ we obtain $|h_1|/h_0 = 2.62 \times 10^{-3}$ with

$N_J \times N_Q = 400 \times 25$ and 2.44×10^{-3} with both 400×100 and 800×25, and the same profile in Fig. 6(a) to within $0.03\,\hat{F}_{max}$. For $\mu = 0.89$ we obtain $|h_1|/h_0 = 0.80$ and $m_c = 0.54$ with 400×25, and 0.85 and 0.52 with 400×100. Close to the transition the function h_1 depends on the number of particles: for $\mu = 0.43$ its mean value is 2.1×10^{-2} with 400×25 and 1.2×10^{-2} with 400×100. This is to be expected, since near the transition the outcome is sensitive to small perturbations, such as those which are due to the discreteness of the particles. However, both simulations lie on the same side of the transition, and the properties of the core agree: $m_c = 0.105$ and 0.103, and the decay times of its oscillation differ by 20 percent. [(Erik) Not tested yet for the cloud with self-gravity: the dependence on Δr (the root mean square radial extension of the core, 0.12 to 0.17, is comparable to $\Delta r = 0.1$), on Δt, and on N_J close to the transition. These tests require new simulations.]

In the Kepler potential the clouds with $M_{gas}/M = 0.01$ and 0.02 ($\mu = 0.26$ and 0.37) relax, which is the behavior described by Ref. [35] for small data, whereas those with $M_{gas}/M = 0.04$ and 0.08 ($\mu = 0.51$ and 0.69) leave an oscillating core. Since μ grows with the mass, for a cloud of a given shape the relaxation fails above a critical mass, which lies between $M_{gas}/M = 0.02$ and 0.04 in this case. Note that the core forms at the circular orbit, which is excluded from the support of the initial data considered in Ref. [35]. [(Erik) A test with a cloud whose support is bounded away from the circular orbit, as in Ref. [35], would show whether a core also forms in that case.]

VII. CONCLUSIONS

In these notes we have studied numerically the relaxation through phase space mixing of a spherically symmetric collisionless gas which is trapped in an external central potential and whose initial DF is concentrated about one orbital phase, taking into account the self-gravity of the gas, in the reduced model in which all the gas particles have the same magnitude of the angular momentum. Our analysis is based on a PIC code whose particles are placed on a regular lattice in action-angle variables. In this section we summarize the results and list some open problems.

In the absence of self-gravity, the PIC code reproduces the exact solution of the mixing problem to within 6×10^{-9} in relative terms over 10^4 time units, during which the first harmonic of the DF decays by seven orders of magnitude. When the self-gravity of the gas is taken into account, the observables which measure the mixing have to be computed with the action-angle variables of the total potential, or of the instantaneous potential when the gas is far from equilibrium. The action-angle variables of the external potential leave a time-independent residue which is proportional to the mass of the gas (2×10^{-3} for $M_{gas}/M = 10^{-3}$) and which could be mistaken for an incomplete mixing.

The self-gravity of the cloud changes its relaxation in three ways, which are controlled by the parameter (50) that compares the amplitude of the potential generated by the cloud with the spread of its orbital frequencies. First, it slows down the relaxation: the potential which the gas generates approaches a stationary profile while the DF continues to mix, which is the analogue of Landau damping, but the time in which its deviation falls below 10^{-3} of its largest value grows from 2.9 units of the mixing time τ_1 without self-gravity to $6.2\,\tau_1$ for $\mu = 0.43$. Second, it changes the state which is reached. In the absence of self-gravity this state is the average of the initial DF over the angle variables. With self-gravity the action variables of the gas particles are not conserved while the potential of the cloud oscillates, and gas accumulates on nearly circular orbits: for $\mu = 0.31$, which corresponds to a cloud with one percent of the central mass, the fraction of the mass with $J < 0.2\sigma_J$ is 6.1 percent instead of 0.5 percent, although the gas is completely mixed. For $\mu \gtrsim 0.4$ a compact part of the cloud is transported to the circular orbit before it is mixed and survives there as a dense core, in which the DF exceeds the largest value that can be obtained by mixing at fixed action; for $\mu = 0.43$ the core contains 10 percent of the mass of the gas, and its radial oscillation is damped with a decay time of about $4\,\tau_1$. Third, above a critical value μ_c the oscillation of the core is not damped during the 30 units of τ_1 covered by our simulations, and the potential does not relax. For clouds which are compact in angle, μ_c lies between 0.43 and 0.48 in the isochrone potential as well as in the Kepler potential, and the simulations with other widths in action are compatible with this interval, whereas for a cloud which is wide in angle the transition lies between $\mu = 0.52$ and $\mu = 0.69$. In the Kepler potential this means that, for a cloud of a given shape, the relaxation by phase mixing described in Ref. [35] only takes place below a critical mass, which lies between 0.02 and 0.04 times the central mass for the clouds considered here.

Several questions remain open. A theory of the critical value μ_c is missing, which should also explain why a cloud which is wide in angle requires a larger value; the heuristic argument of section VI A only shows that μ_c should be of order one. The mechanism which damps the oscillation of the core below μ_c and sustains it above, and its relation to the trapping of resonant particles in plasma waves, remains to be analyzed. Except for the largest masses, our simulations do not show whether the core is held together by its own gravity. The dependence of the results on the grid spacing, on the time step and on the number of rows of the lattice close to the transition has not been tested, and the decay laws of $|h_1|$ and of the potential require simulations with more particles than those used here. It is not known whether the oscillation of the core above μ_c is eventually damped, as in the core-halo states of systems with long-range interactions [36], and on which time scale; answering this question requires longer simulations. Finally, all

our results refer to the reduced model, and the core forms at the circular orbit, which is excluded from the support of the data considered in Ref. [35]. When the angular momenta of the gas particles are distributed, the frequencies of the orbits depend also on L, which provides an additional source of mixing (see section III), and the initial data can be chosen to satisfy the assumptions of that reference. Whether a core forms in that case, and whether the critical value of μ is shifted, is a natural next step, for which the methods of these notes can be extended in a direct way. [(Erik) To be completed after the discussion with Olivier: the outlook regarding the distribution of angular momenta, and the reference to the companion article on perturbations of steady states.]

Acknowledgments

[(Erik) To be written.]

Appendix A: Numerical computation of the action-angle variables

In this appendix we describe the numerical evaluation of the action-angle variables defined in Eqs. (16,17,18) for a potential of the form $\Phi = \Phi_{ext} + \Phi_{gas}$, in which Φ_{gas} is only known on a radial grid. This is required for the construction of the self-consistent steady states, for the generation of the initial data and for the computation of the observables $h_n(t)$.

The values of Φ_{gas} on the grid are interpolated with a natural cubic spline, and beyond the last grid point we set $\Phi_{gas} = -GM_{gas}/r$. Given a point (r, p_r, L) we compute its energy E from Eq. (15), locate the minimum $r_{min}(L)$ of the effective potential V_L, and determine the turning points $r_1 < r_{min}(L) < r_2$ by solving the equation $V_L(r) = E$ with the bisection method on either side of the minimum. The integrand in Eq. (16) behaves like a square root at the turning points and the one in Eq. (17) like an inverse square root, which spoils the convergence of standard quadrature rules. We avoid this problem by means of the substitution

$$r = r_m + r_a \sin\theta, \qquad r_m := \frac{r_1 + r_2}{2}, \quad r_a := \frac{r_2 - r_1}{2}, \qquad -\frac{\pi}{2} \le \theta \le \frac{\pi}{2}. \tag{A1}$$

Since for a non-circular orbit the zeros of the function $E - V_L$ at the turning points are simple, one has $E - V_L(r) = r_a^2 \cos^2\theta\, u(\theta)$ with a smooth and positive function u, and it follows that

$$A(E,L) = 2r_a^2 \int_{-\pi/2}^{\pi/2} \cos^2\theta \sqrt{2u(\theta)}\, d\theta, \qquad T(E,L) = 2 \int_{-\pi/2}^{\pi/2} \frac{d\theta}{\sqrt{2u(\theta)}}. \tag{A2}$$

Both integrands are smooth, and we evaluate the integrals with a Gauss-Legendre rule with 64 nodes. The angle variable is obtained in the same way from

$$Q = \frac{2\pi}{T(E,L)} \int_{-\pi/2}^{\theta(r)} \frac{d\theta'}{\sqrt{2u(\theta')}}, \qquad \theta(r) := \arcsin\left(\frac{r - r_m}{r_a}\right), \tag{A3}$$

when $p_r \ge 0$, and from 2π minus the right-hand side of Eq. (A3) when $p_r < 0$. For the isochrone potential without self-gravity, for which the exact expressions (22,23) are available, this procedure yields the action variable with an error of the order of 10^{-14} and the period with an error of the order of 10^{-11}. Without the substitution (A1) the corresponding errors are 2×10^{-6} and 9×10^{-3}, and they only decrease like the third and the first inverse power of the number of nodes, respectively.

For the inverse map $(Q, J) \mapsto (r, p_r)$ at fixed L we tabulate the function $J = A(E,L)/2\pi$ for 4000 values of the energy between $E_{min}(L)$ and a value slightly above the cutoff energy, and obtain $E(J,L)$ by interpolation. Next, we determine the turning points and find, by bisection in θ, the point of the orbit at which the time elapsed since the passage through the pericenter is equal to $QT(E,L)/2\pi$ when $Q \le \pi$ and to $(2\pi - Q)T(E,L)/2\pi$ otherwise. This gives the radius r, and the momentum is $p_r = \pm\sqrt{2(E - V_L(r))}$, with the positive sign when $Q \le \pi$. Applying the direct map to the result reproduces the points of the lattice (41) with errors which are smaller than 10^{-8} in J and 5×10^{-7} in Q. The frequency is obtained from $\Omega = 2\pi/T(E,L)$.

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